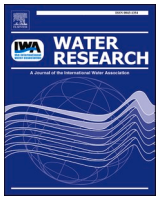
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Coupling of membrane-based bubbleless micro-aeration for 2,4-dinitrophenol degradation in a hydrolysis acidification reactor

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ABSTRACT

Micro-aeration hydrolysis acidification (HA) is an effective method to enhance the removal of toxic and refractory organic matter, but the difficulty in stable dosing control of trace oxygen limits its wide application. Membrane-based bubbleless aeration has been proved as an ideal aeration method because of its higher oxygen transfer rate, more uniform mass transfer, and lower cost than HA. However, the available information on its application in HA is limited. In this study, membrane-based bubbleless micro-aeration coupled with hydrolysis acidification (MBL-MHA) was exploited to investigate the performance of 2,4-dinitrophenol (2,4-DNP) degradation via comparing it with bubble micro-aeration HA (MHA) and anaerobic HA. The results indicated that the performances in MBL-MHA and MHA were higher than those in HA during the experiment. 2,4-DNP degradation rates under redox microenvironments caused by counter-diffusion in MBL-MHA (84.43~97.28%) were higher than those caused by co-diffusion in MHA (82.41~94.71%) under micro-aeration of 0.5–5.0 mL air/min. The 2,4-DNP degradation pathways in MBL-MHA were nitroreduction, deamination, aromatic ring cleavage, and fermentation, while those in MHA were hydroxylation, aromatic ring cleavage, and fermentation. Reduction/oxidation-related, interspecific electron transfer-related species, and fermentative species in MBL-MHA were more abundant than that in MHA. Ultimately, more reducing/oxidizing forces formed by more redox proteins/enzymes from these rich species could enhance 2,4-DNP degradation in MBL-MHA.

1. Introduction

Nitrophenols have been extensively used as essential raw materials for dyes, plasticizers, pesticides, herbicides, paints, pharmaceuticals, explosives, and leather preservatives in the global chemical industry (Wu et al., 2015; Fajardo et al., 2015; Arora et al., 2014). Among the nitrophenols, 2,4-dinitrophenol (2,4-DNP), a representative nitrophenol, has been ranked as a priority-controlled pollutant by the United States Environmental Protection Agency (2003) and must be appropriately treated because of its high toxicity, oncogenicity, and biological accumulation (Sasidharan Pillai et al., 2015; Zhang et al., 2021).

Several methods to treat 2,4-DNP-containing wastewater via physicochemical treatment processes, for example, adsorption methods (Azari et al., 2021), advanced reduction (Yang et al., 2011), advanced oxidization (Hedayat et al., 2018), and various other methods (Xiong et al., 2019), have been investigated in the literature. Although they have achieved remarkable effects, these methods have disadvantages, such as the necessity to recycle chemicals, the complex processes involved, the high cost, and the resultant secondary pollution (Naidu et al., 2016). Therefore, in recent decades, biological treatment methods have been widely used because of their high degradation efficiency and uncomplicated treatment or extraction processes.

Hydrolytic acidification (HA) is an effective, economical biotreatment method that plays a crucial role in changing refractory complex macromolecules, such as aromatic hydrocarbons or heterocyclic substances, into small molecules, such as small-molecule organic acids and alcohols, and improving biodegradability via biological hydrolysis and acid production (G Song et al., 2017; Zhang et al., 2021). Therefore,

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HA has been widely used to treat wastewater containing refractory and toxic organic matter (G Song et al., 2017). However, the hydrolysis of complex organic matter in the HA process is usually a rate-limiting step (Song et al., 2019; Wu et al., 2019).

To solve this problem, the literature has proposed new methods for enhancing hydrolysis efficiencies. Recently, studies have reported that micro-aeration (dosing tract air or oxygen, also known as limited aeration) is an effective method to accelerate the hydrolysis of complex organic matter, improve volatile fatty acid (VFA) production, and increase biodegradability (Nguyen et al., 2019; Song et al., 2019; Zhang et al., 2021). In general, a dissolved oxygen (DO) concentration in the range of 0.1–0.5 mg/L, corresponding to the oxidation–reduction potential (ORP) value of −300–0 mV, can be defined as a micro-aeration environment (Wu et al., 2019; Nguyen et al., 2019; Zhang et al., 2021). Studies have reported that micro-aeration HA enhances the hydrolysis of various refractory organic matter in wastewater, such as nitrophenol, benzene, toluene, ethylbenzene, xylene (BTEX), aniline, and aldehydes compounds (Cheng et al., 2015; Firmino et al., 2017; João et al., 2018; Khan et al., 2019; Song et al., 2019). For example, Khan et al. (2019) reported that under 4-nitrophenol (4-NP) of 100 mg/L in the influent, the 4-NP degradation rate reached 99.89% under micro-aeration conditions (DO = 0.5 mg/L), and it was only 4% under anaerobic conditions (DO = 0 mg/L). João et al. (2018) demonstrated that under the conditions from the anaerobic to the micro-aeration of 1.0 mL air/min, the removal rates of benzene and toluene increased from 55% to 67% and from 84% to 89%, respectively. The aniline removal rate reached up to 91.2 ± 2.2% under micro-aeration of 0.31 ± 0.17 mg/L, and it was 15.8 ± 1.1% without micro-aeration (Cheng et al., 2015). The phenol removal rate reached 80.7% in the micro-aerated condition (DO = 0.41 mg/L) and was 11.6% under anaerobic conditions (Yang et al., 2018). Nevertheless, the stable dosing and precise control of trace oxygen is difficult, which has limited the application of limited aeration as an effective operational strategy in HA (Kong et al., 2019; Nguyen et al., 2019). For the stable dosing of trace oxygen in HA, a suitable micro-aeration dosing method should be established because it could affect the oxygen transfer rate (OTR) and oxygen transfer efficiency by affecting the bubble size (Zhang et al., 2021). Micro-aeration dosing methods usually include limited aeration, hydrogen peroxide (H_2O_2), and electrolysis (Wu et al., 2018; Tartakovsky et al., 2003; 2011). Although some chemical oxidants, such as H_2O_2 and calcium peroxide, can release peroxides by reacting with water to elevate O_2 levels, they can also kill microorganisms and pollute aquatic ecosystems (D. Wang et al., 2019). Limited aeration would be more promising than chemical oxidants because it could be nontoxic and without chemical additives, especially for the enhancement of the HA process, to fulfill the sustainable development goals. Generally, under limited aeration, the smaller bubble size, the higher OTR and the lower costs, resulting from higher interfacial areas, lower rise velocities, stronger stability, and higher gas utilization (H. Wang et al., 2020; John et al., 2020). Temesgen et al. (2017) reported that the OTR of nanobubbles (diameter <1 μm) was 125 times faster than that of bulk bubbles (diameter >1000 nm). John et al. (2020) investigated the effect of limited aeration with nanobubbles and bulk bubbles into HA on cellulose degradation. The results demonstrated that under the nanobubble micro-aeration addition, the reductions in cellulose content and cellulose crystallinity increased by 8%–14% and 9%–21%, respectively, in comparison with the bulk bubble micro-aeration.

In recent years, hydrophobic membranes, such as silicone tubes, polyvinylidene fluoride membranes, and polypropylene membranes, have proved ideal for injecting oxygen for efficient gas transfer. Unlike bubble aeration, pressurized gas diffuses into the solution through the membrane module without forming bubbles, achieving an approximately 100% higher OTR, more uniform mass transfer, a lower supply pressure, and a lower cost than bubble aeration (Aybar et al., 2014; Perez-Calleja et al., 2017; Hou et al., 2019). The membranes can also serve as microorganism carriers, facilitate abundant microorganisms to colonize the rough membrane surface, and form biofilms (Kinh et al., 2017). These biofilms consist of a stratified community embedded within a tight self-produced matrix of extracellular polymeric substances (EPS) (Pellicer-Nacher et al., 2014), which are highly resistant to toxicity and load shock (Li et al., 2015). Membrane-based bubbleless aeration as an oxygen dosing method has been applied in activated sludge systems and mainly focused on simultaneous nitrification and denitrification, hydrogen sulfide, and organic matter removal (Wang et al., 2009; Sun et al., 2016; X. Mei et al., 2019). However, the literature is limited on the coupling of the membrane-based bubbleless micro-aeration in HA for degrading toxic and refractory organic matters to nontoxic or low-toxicity small molecular organic matters and ensuring the subsequent biodegradability, rather than for removing chemical oxygen demand (COD). What might be feasible is to create a more stable and uniform micro-aeration environment through the coupling of membrane-based bubbleless micro-aeration in HA; however, few attempts have been reported. Notably, under the membrane-based bubbleless micro-aeration addition, the supplied oxygen diffusion could be from the base of the biofilm to the bulk liquid, and the supplied substrate diffusion could be from the bulk liquid to the base of the biofilm (counter-diffusion). Under the bubble micro-aeration addition, both supplied oxygen and substrate diffusion could be from the bulk liquid to the surface of the suspended sludge (co-diffusion) (Wang et al., 2009). As a result, the microbial community structure could have different spatial microstructures between the two aeration methods, caused by counter- and co-diffusion, such as aerobic, anoxic, and anaerobic zones from the base of the biofilm to the bulk liquid for bubbleless aeration (X. Mei et al., 2019), and anaerobic, anoxic, and aerobic zones from the center of the suspended sludge to the bulk liquid for bubble aeration (Nguyen et al., 2019).

This opposite DO spatial microstructure caused by counter- and co-diffusion might lead to different pollutant degradation effects. According to the literature, degradation products produced by 2,4-DNP differ under anaerobic and micro-aeration conditions; for example, under anaerobic conditions, degradation products of 2,4-DNP have been proven to be corresponding amino phenols, namely, 2,4-diaminophenol (2,4-DAP), 2-amino-4-nitrophenol (2-A-4-NP), and 2-nitro-4-aminophenol (2-N-4-AP) (Kimura et al., 2003; Tang et al., 2018), and under oxygen-containing conditions, nitro groups of 2,4-DNP could be released from the benzene ring, and benzenethiol could be generated, in which oxygenases play a major role (X. Mei et al., 2019). Hence, the degradation effects of 2,4-DNP might be different using the bubble and membrane-based bubbleless limited aeration into HA due to the opposite DO spatial microstructure. However, to date, the effects of bubble and membrane-based bubbleless micro-aeration on 2,4-DNP degradation in HA have rarely been documented. Limited insights into the effects of limited aeration on the degradation pathways of 2,4-DNP and the molecular mechanisms of microbial community and microbiome protein expression have been reported thus far.

Therefore, in this study, a membrane-based bubbleless micro-aeration coupled with the HA (MBL-MHA) process was exploited for 2,4-DNP degradation. For the precise control of micro-aeration, MBL-MHA was equipped with an automatic temperature control system and a DO online automatic monitoring system, which was integrated into a programmable logic controller (PLC) system. It was operated continuously for 100 days to investigate the performance of 2,4-DNP degradation by comparing it with that of bubble micro-aeration HA (MHA) and anaerobic HA, and the objectives of this study were to (i) evaluate the performance of 2,4-DNP degradation in MBL-MHA and MHA under different limited aeration rates; (ii) investigate the degradation pathways of 2,4-DNP in the MBL-MHA and MHA systems; and (iii) reveal the synergistic effects of 2,4-DNP degradation, the microbial community, and microbiomes' protein expression in MBL-MHA and MHA.

2. Materials and methods

2.1. Simulated wastewater and inoculum

The three bioreactors were fed simulated wastewater. The specific characteristics of the simulated wastewater were found as follows: NH_4Cl of 191.4 mg/L, KH_2PO_4 of 60 mg/L, $MgCl_2 \bullet 6H_2O$ of 200 mg/L, $FeSO_4 \bullet 7H_2O$ of 25 mg/L, $CuSO_4 \bullet 5H_2O$ of 5 mg/L, $CoCl_2 \bullet 5H_2O$ of 0.125 mg/L, $MnSO_4 \bullet 6H_2O$ of 15 mg/L, and $NaHCO_3$ of 1800 mg/L (Zhang et al., 2019) and pH was 7.1–7.4. Generally, the 2,4-DNP concentration was 2~120 mg/L in actual industrial wastewater (Wu et al., 2015; Chen et al., 2013). Therefore, glucose was used as the primary organic carbon source at a COD concentration of approximately 2000 mg/L. The 2, 4-DNP concentration was 20.0 mg/L in the simulated wastewater, corresponding to a COD concentration of approximately 51.0 mg/L. Total nitrogen (TN) was approximately 55 ± 1.8 mg/L, and UV_{254} was 0.85 ± 0.03 in the simulated wastewater. 2,4-DNP was purchased from Sigma-Aldrich Co. (USA). Other chemicals, such as glucose, $NaHCO_3$, NH_4Cl, KH_2PO_4, $MgSO_4 \bullet 7H_2O$, $FeSO_4 \bullet 7H_2O$, $CuSO_4 \bullet 5H_2O$, $CoCl_2 \bullet 5H_2O$, and $MnSO_4 \bullet 6H_2O$, were obtained from the Xihui Chemical Factory (Chaoyang, Beijing). 2,4-DNP was guaranteed reagent grade, and the other chemicals were of analytical reagent grade.

The inoculum sludge was collected from a 100 L expanded granular sludge bed reactor that has stably operated for more than 2 years with glucose as the organic substrate and an organic load rate of 6.0 kg COD/m^3 d in the laboratory. The concentration of suspended solids (SS) and volatile suspended solids (VSS) was 12.5 ± 0.5 g/L and 9.2 ± 0.4 g/L, and VSS/SS was approximately 0.73. Characteristics of the inoculum sludge are listed in Tab. S1.

2.2. Bioreactor set-up and operation

In this experiment, three HA reactors were operated in parallel: MBL-MHA, MHA, and HA as a control operated under anaerobic conditions. A hollow fiber membrane made from polyvinylidene fluoride (PVDF) for bubbleless micro-aeration was supplied by MBL-MHA. The membrane module was composed of ten membrane tubes with a length of 100 cm and was encapsulated by epoxy resin. It had an inner diameter of 2.5 mm, an outer diameter of 2.8 mm, a wall thickness of 0.15 mm, a micropore size of 0.15 μm, and a contact angle of 89.84 ± 1.57° The bubble micro-aerator, located at the bottom of the MHA, was made from titanium alloy and had a diameter of 50 mm and micropore size of 5 μm. Each reactor was made of acrylic plexiglass with an internal diameter of 50 mm and a total height of 1200 mm and had a total working volume of 2.35 L. The reactors were operated with a hydraulic retention time (HRT) of 24 h. A schematic of these reactors is shown in Fig. 1.

An automatic temperature control system was installed in each reactor to maintain the temperature of the system at 25 ± 0.2 °C. The air flow rate from the compressor tank was finely maintained using a trace mass gas flowmeter (Horiba S49 33/MT, China). The DO concentration was measured using an automatically monitored online system (INESA SJG-9435B, China). To avoid the DO probe surface by air or oxygen bubble adsorption, which affects the monitoring accuracy, DO was monitored at the ectopic site in these bioreactors. The DO monitoring probe was outside the reactor, and the mixed liquid in the reactor flowed through the DO monitoring probe in the form of uniform turbulent flow to achieve DO monitoring. The automatic temperature control system, trace mass gas flowmeter, and DO were automatically monitored online and controlled using a PLC system (SIEMENS, Germany).

The specific process was as follows: The DO online monitoring probe was immersed in the flow cup. The mixture in the reactor continuously and stably flowed into the flow cup from the bottom through a peristaltic pump and then flowed out to the secondary sedimentation tank through overflow in the flow cup. The entire process was conducted under sealed conditions. A schematic of the circulation cup structure is shown in Fig. S1. When the air pressure was below the membrane bubble point pressure, its bubbleless aeration could form, achieving an approximately 100% higher OTR, more uniform mass transfer, a lower supply pressure,

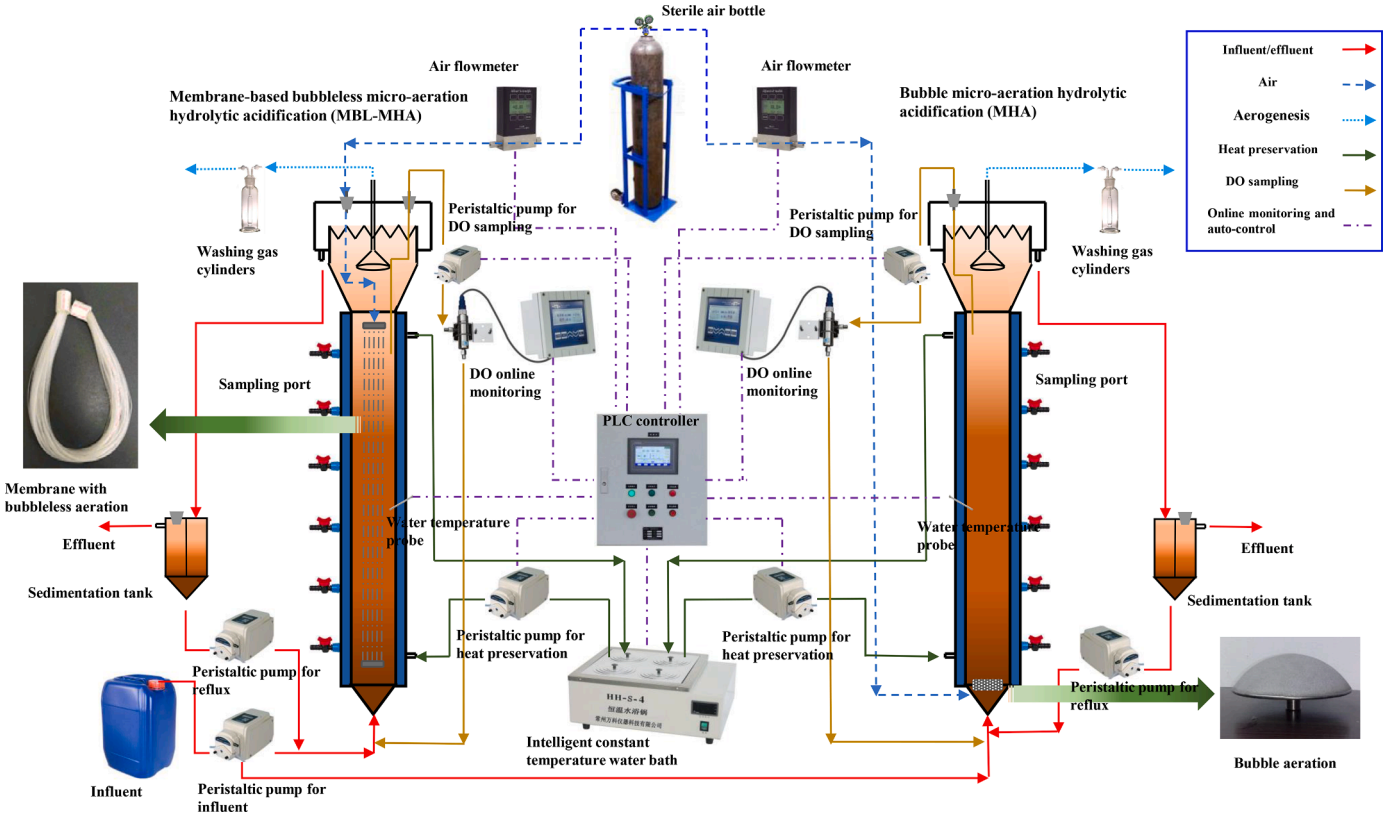


Fig. 1. Schematic diagram of the micro-aeration HA process for 2,4-DNP degradation.

and a lower cost than bubble aeration (Aybar et al., 2014; Perez-Calleja et al., 2017; Hou et al., 2019). It is therefore necessary to determine the membrane bubble point pressure. The specific determination steps were based on those in the literature (Wu et al., 2019). The membrane module was completely immersed in water with one end connected to a sterile pressurized air bottle and the other end sealed. Then a trace mass gas flowmeter was switched on to control the air flow rate. The bubble point was the pressure in the membrane's inner lumen when the first bubble began to appear on the membrane's surface.

The performance of 2,4-DNP degradation in the three bioreactors under different limited aeration conditions was investigated. The operating conditions of these bioreactors comprised three phases: phase I (air flow rate of 0.5 mL/min, days 1–35), phase II (air flow rate of 3.0 mL/min, days 36–65), and phase III (air flow rate of 5.0 mL/min, days 66–100). The air pressure was 25 kPa at each phase. Each phase was operated for no less than 30 days to ensure the stable operation of each phase. In each stage, stable performance (a quasi-steady state) was achieved (no significant difference [ANOVA, $p > 0.05$] in each monitoring parameter during five consecutive days of operation), and the next stage was operated. The pH values in each HA during the experiment was 6.4–7.2 (Fig. S2).

2.3. Batch experiment

The nitrogen transformation during 2,4-DNP degradation within a given HRT was investigated, namely, a batch experiment was performed using simulated wastewater without NH_4Cl after the continuous inflow experiment in phase II on day 65. According to the literature (X. Mei et al., 2019), the specific operation process was as follows: First, both reactors were drained empty. Second, the simulated wastewater containing 2,4-DNP was quickly infused within 0.5 min. The initial 2,4-DNP concentration was 20 mg/L. During the batch experiment, the peristaltic pump for the influent was closed. Intermittent sampling and analysis of water quality, including ammonia (NH_4^+-N), nitrite (NO_2^--N), and nitrate (NO_3^--N), were conducted during the batch experiment. The experiments and measurements were performed in triplicate, and the average values were used for the analysis.

2.4. Analytical methods

2.4.1. Water quality analysis

The COD, NH_4^+-N, and TN were analyzed according to standard methods (APHA, 2005). NO_2^--N and NO_3^--N were measured using ion chromatography (Dionex, ICS-1000, Japan). The 2,4-DNP concentration was measured using a high-performance liquid chromatography system (Agilent 1100, USA) equipped with a UV–*vis* detector and an automatic sample injection device with a loop volume of 10 mL (Liang et al., 2012). The intermediate products of 2,4-DNP degradation were measured using a liquid chromatography-mass spectrometer (LC-MS) system (Waters, Waters2695, America). The detailed analysis process is described in Henriques et al. (2007). The pH and ORP in the reactors were measured using a portable meter (HACH, HQ11d, America). DO and ORP in the sludge were detected using a microelectrode instrument (Unisense, Denmark). VFAs were determined using a gas chromatograph equipped with a flame ionization detector (Agilent 7890, USA). The detailed analysis process was in accordance with the literature (G Song et al., 2017). The oxygen transfer test of the bubble and membrane-based bubbleless micro-aeration was performed. Detailed information is provided in Supporting Information A Text S1. Biomass of the biofilm on the membrane surface was calculated utilizing the gravimetric method. The detailed analysis process was in accordance with the literature (Wu et al., 2019). Detailed information is provided in Supporting Information A Text S2. Measurements were performed in triplicate, and average values were used for analysis.

Acidification degree (AD) is an essential parameter for evaluating the efficiency of fermentation. The AD for each operation phase was calculated using Eq. (1).

$$AD = \frac{\Delta C_f}{C_i} \times 100 \quad (1)$$

where C_i is the soluble COD of the influent (mg/L), and Δ and C_f produce net VFAs (final initial), expressed as theoretical equivalents of COD as mg/L. The COD equivalents of each VFA were as follows: acetic acid, 1.07; propionic acid, 1.51; butyric acid, 1.82; 2-methyl butyric acid, 2.04; valeric acid, 2.04; 2-ethyl butyric, 2.21; and hexanoic acid, 2.21 (G Song et al., 2017).

2.4.2. EPS extraction and analysis

EPS, including loosely bound EPS (LB-EPS) and tightly bound EPS (TB-EPS), were extracted from the activated sludge in MBL-MHA, MHA, and anaerobic HA under stable operation conditions using a heat extraction method. The specific extraction steps were based on those in the literature (Guo et al., 2020; Zhang et al., 2021). The contents of proteins, polysaccharides, and nucleic acids in the LB-EPS and TB-EPS were also measured. Proteins were detected using a BCA protein kit (Sangon, China), polysaccharides were measured using the phenol-sulfuric acid method, and nucleic acids were evaluated using a TIANamp Bacteria DNA Kit (Tiangen, China) (Song et al., 2019).

The FT-IR spectra of freeze-dried LB-EPS and TB-EPS in potassium bromide (KBr) pellets were determined using an FT-IR spectrometer (Nicolet 6700, Thermo Scientific Co., USA). The resolution was 4 cm^{-1}, and the scan range was 400–4000 cm^{-1}. The freeze-drying of LB-EPS and TB-EPS was conducted using a lyophilizer (FD-1A-50, Beijing Boyikang Laboratory Instruments Co., Ltd., China).

The UV–*vis* absorption spectra of LB-EPS and TB-EPS were obtained using an automatic scanning UV–*vis* spectrophotometer (TU-1810, Beijing Purkinje General Instrument Co., Ltd., China) at 200–400 nm.

The 3D excitation emission matrix (3D-EEM) fluorescence spectra of LB-EPS and TB-EPS were obtained using a fluorescence spectrum analyzer (F-7000, Hitachi, Japan). The excitation (Ex) and emission (Em) wavelength ranges were 200–550 nm and 280–550 nm, respectively. The excitation and emission slit widths were adjusted to 5 nm, and the scan speed was 1200 nm/min (Zhang et al., 2021). The fluorescence peaks of 3D-EEM can be categorized into five main regions: tyrosine-like, tryptophan-like, fulvic acid-like, soluble microbial by-product-like, and humic acid-like (Tab. S2) (Chen et al., 2003; Henderson et al., 2009). The parallel factor (PARAFAC) technique was used to separate the fluorescence spectra from the 3D-EEM to avoid any overlapping of the fluorescence spectra. The specific steps of the PARAFAC analysis are described in Song et al. (2019).

The EPS structure could affect the morphologies of sludge and biofilm (Chen et al., 2021). A scanning electron microscope (SEM) (S4800-HSD, Hitachi, Japan) was used to observe the microstructure of suspended sludge of MBL-MHA, biofilm on the membrane surface of MBL-MHA, and suspended sludges of MHA and anaerobic HA during stable operation in phase II.

2.4.3. DNA extraction and high-throughput sequencing analysis

In this study, sludge samples, including suspended sludge of MBL-MHA (MBL-MHA-I), biofilm on the membrane surface of MBL-MHA (MBL-MHA-II), and suspended sludges of MHA and anaerobic HA, were collected during stable operation in phase II. The operation process of microbial structure analysis was based on the literature (Song et al., 2019; Tian et al., 2020). DNA was extracted from sludge samples using the Tiangen® soil DNA kit (Tiangen Biotech, China) by following the manufacturer's protocol. The extracts were stored at −20 °C until further analysis. The V3-V4 regions of the 16 s rDNA genes were amplified with 340 F (CCTACGGGNBGCASCAG) and 805R (GACTACNVGGGTATCTAATCC). All polymerase chain reaction mixtures were purified using MinElute Gel Extraction Kit (Qiagen, Germany). The Invitrogen™-Qubit fluorometer quantitative system (California, USA)

was used to quantify the amplicons. High-throughput sequencing of 16S DNA amplicons was performed using the Illumina MiSeq platform at SinoGenoMax Co., Ltd (Beijing, China).

2.5. Label-free quantitative proteomic profiling

Four samples—MBL-MHA-I, MBL-MHA-II, MHA, and anaerobic HA—were selected for further analysis by label-free quantitative proteomics. The operation process referred to the literature (Zhang et al., 2019, 2020). The detailed information is provided in the Supporting Information A Text S3.

3. Results and discussion

3.1. Performances of bioreactors under different air flow rates

During each phase, MBL-MHA, MHA, and anaerobic HA (air flow rate of 0 mL/min) were gradually stabilized after continuous running for one month. These reactors were run continuously for 100 days from phase I to phase III to investigate the performance of 2,4-DNP degradation in MBL-MHA and MHA by setting the micro-aeration rate from 0 to 5.0 mL/min (Fig. 2).

As shown in Fig. 2, in phase I, the removal rates of 2,4-DNP, COD, TN, and UV_{254} in the micro-aeration HA (MBL-MHA and MHA) at an air flow rate of 0.5 mL/min were obviously higher than that in the anaerobic HA. AD and effluent OUR were significantly increased in MBL-MHA and MHA but not in anaerobic HA. These results indicate that limited aeration into HA could result in an enhanced 2,4-DNP removal rate and acid production. Additionally, with an increasing air flow rate from 0.5 to 3.0 mL/min, the 2,4-DNP removal rates in MBL-MHA and MHA ranged from 84.32 ± 1.90% to 97.28 ± 0.51%, and from 82.41 ± 0.43% to 94.71 ± 0.35%, respectively. The COD and TN removal rates also increased, from 55.42 ± 1.11% to 68.13 ± 1.24% and from 42.21 ± 1.11% to 51.87 ± 0.87% in MBL-MHA, from 54.31 ± 0.46% to 65.32 ± 0.94% and from 39.43 ± 0.35% to 45.69 ± 0.73% in MHA, respectively. When the air flow rate was from 0.5 to 3.0 mL/min, under the UV_{254} value of 0.65 ± 0.02 in the influent, UV_{254} values in the effluent also obviously decreased, from 0.12 ± 0.02 to 0.04 ± 0.01 in MBL-MHA and from 0.15 ± 0.02 to 0.07 ± 0.01 in MHA. The lower UV_{254} values, the lower aromatic constituents in the wastewater (Wan et al., 2020).

These results demonstrated that at an air flow rate of 3.0 L/min, the aromatic constituent in the effluent of the micro-aeration HA effluent was lower than that at an air flow rate of 0.5 mL/min. The high aromatic constituent (UV_{254} value) in the influent was mainly contributed to the 2,4-DNP existed in the influent. In the two micro-aeration HA systems at an air flow rate of 3.0 L/min, 2,4-DNP degradation rates (97.28 ± 0.51% of MBL-MHA and 94.71 ± 0.35% of MHA) were higher than the UV_{254} removal rates (93.85 ± 0.18% of MBL-MHA and 89.23 ± 0.21% of MHA), indicating that the degradation intermediates of 2,4-DNP might contain aromatic organics and could mainly contribute to the effluent aromatic constituent. As a result, the lower UV_{254} values in the effluent of the two micro-aeration HA systems at an air flow rate of 3.0 L/min were compared with those of 0.5 mL/min and indicated that the content of aromatic organics from the degradation intermediates of 2,4-DNP of 3.0 L/min was lower than that of 0.5 mL/min and that 2,4-DNP could be degraded even more thoroughly than in the literature. Additionally, effluent AD at an air flow rate of 3.0 mL/min was significantly increased to 78.48 ± 1.51% in MBL-MHA and 74.58 ± 1.72% in MHA, and it was 73.21 ± 1.33% in MBL-MHA and 68.73 ± 1.51% in MHA at an air flow rate of 0.5 mL/min, indicating that limited aeration (3.0 mL/min) could improve VFA production. VFAs can serve as organic carbon sources to accelerate the removal of biological nutrients and improve the biodegradability of wastewater (W.S. Lee et al., 2014). Additionally, under an air flow rate from 0.5 mL/min to 3.0 mL/min, effluent OUR values also obviously improved, from 14.82 ± 0.41 mg O_2/g-MLSS-h to 18.42 ± 0.31 mg O_2/g-MLSS-h in MBL-MHA and from 13.34 ± 0.53 mg O_2/g-MLSS-h to 15.84 ± 0.52 mg O_2/g-MLSS-h in MHA, respectively. These results indicated that the inhibitory effect of the influent containing toxic 2,4-DNP on activated sludge decreased obviously through the treatment of micro-aeration HA. The limited aeration (3.0 mL/min) into HA can enhance the treatability of the wastewater containing toxic 2,4-DNP, and the effluent is more suitable for the following aerobic biological treatment processes. Firmo et al. (2018) reported that under the BTEX of 3 mg/L in the influent, when the air flow rate was 0.5 mL/min, the micro-aeration HA assured high removal efficiencies (> 80%) for all compounds of BTEX. João et al. (2018) reported that benzene exhibited a 30% increase in its removal efficiency under a flow rate of 1.0 mL air/min in the micro-aeration HA. A trace amount of oxygen in HA could increase the number of oxygenases to improve the removal of aromatic hydrocarbon compounds (e.g., BTEX and nitrophenol) (Wu et al., 2015; 2018). In this study, the optimum air flow rate was 3.0 mL/min, which was inconsistent with the literature. This result might be

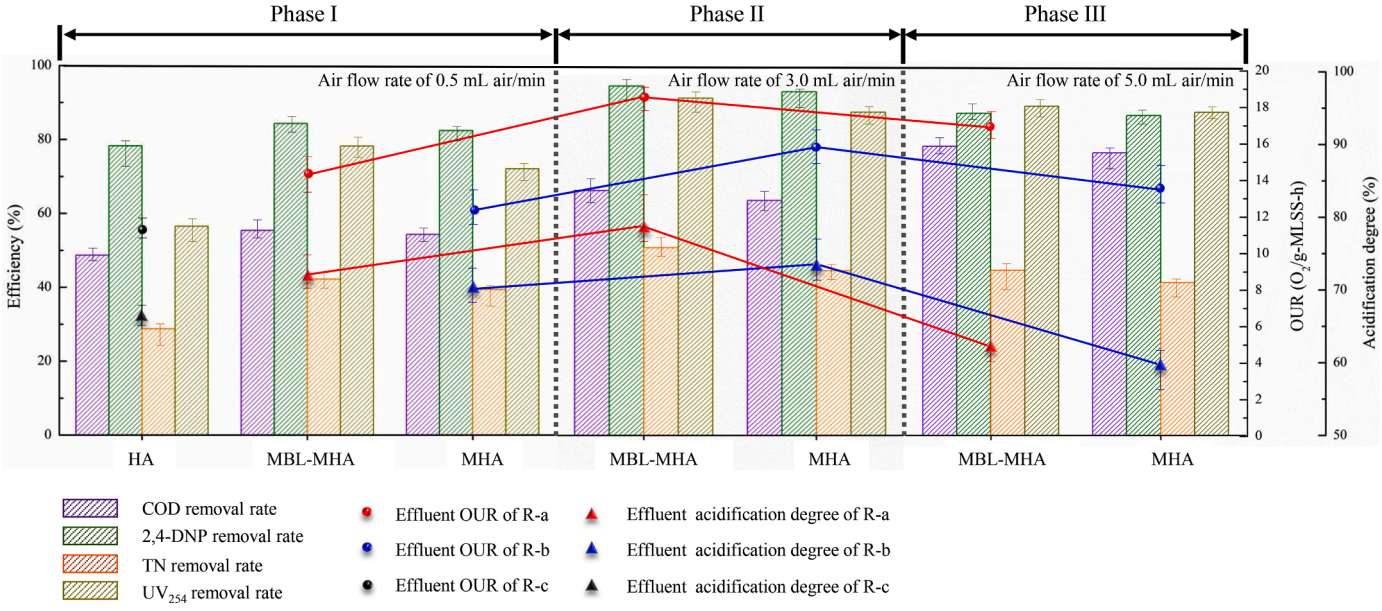


Fig. 2. Performances of the MBL-MHA (R-a), MHA (R-b) under different air flow rates and HA (R-c).

attributed to the optimum DO concentration being related to the type and concentration of the substance. Hence, the effects of organic load rates on the performance of MBL-MHA for the treatment of 2,4-DNP should be investigated.

However, with a further increase in the air flow rate to 5.0 mL/min, the 2,4-DNP, TN, and UV_{254} removal rates and effluent OUR decreased in MBL-MHA and MHA. The deterioration of 2,4-DNP degradation when applying more air could be attributed to the following: under excessive oxygen, 2,4-DNP could not be easily degraded because of its chemical structures with electron-absorbing nitro groups and negatively charged benzene rings (Zhang et al., 2021). In addition, the biomass of the biofilm on the membrane surface in phase I-III were 20.98±0.46 g/m^2, 25.85±0.29 g/m^2, and 23.91±0.62 g/m^2, which demonstrated that the appropriate air flow rate could promote the growth of microorganisms on the membrane surface, while excessive oxygen might destroy the micro-aeration environment and reduce the biomass on the membrane surface. Notably, AD decreased to 61.45 ± 2.32% in MBL-MHA and 58.32 ± 1.28% in MHA, and the COD removal rates obviously increased to 78.36 ± 1.45% in MBL-MHA and 76.82 ± 1.12% in MHA. Similar results have been reported (Mozo et al., 2011; Wu et al., 2015). For example, Wu et al. (2015) reported that with a further increase in DO from 0.3 to 0.4 mg/L to 0.4–0.5 mg/L for the pretreatment of petrochemical wastewater in HA, VFA production decreased from 1.68 mg/L to 1.01 mg/L, the BOD_5/COD ratio of the effluent decreased significantly from 0.31 to 0.20, and the COD removal rate increased from 24.9% to 31.6%. These results were mainly attributed to the following: VFAs can be oxidized by excess oxygen when applying more air. The 2,4-DNP degradation and fermentation ability could be enhanced in the micro-aeration HA system when applying a suitable limited aeration (3.0 mL/min) into the HA; however, a much higher limited aeration would result in a negative effect on reactor performance in the micro-aeration HA.

In this study, 2,4-DNP degradation efficiency using the micro-aeration HA at phase II was higher than the reports in the literature for 2,4-DNP degradation using an anaerobic biological process; for example, Dai et al. (2009) reported that under 2,4-DNP of 0.25 mmol/L in wastewater, the 2,4-DNP degradation rate was 75.28% after adding NADH in an anaerobic biological process. She et al. (2005) reported that under 2,4-DNP of 2~45 mg/L in wastewater, the 2,4-DNP degradation rate was 10%~65.1% in an anaerobic digestion process. As indicated in the literature, in the presence of low oxygen concentrations, microbial activity can be improved (Nguyen et al., 2019), and the amount of mono-oxygenases can be increased to improve the removal of aromatic hydrocarbon compounds (e.g., nitrophenol) (Varjani, 2017). Additionally, the addition of trace oxygen in an HA as a terminal electron acceptor could accelerate the hydrolysis of long-chain organic compounds under facultative bacteria, promoting 2,4-DNP transformation (Zhang et al., 2021). However, when the applied limited aeration was too high, such as 5.0 mL/min in this study, the oxygen might be excessive in the reactor, which could cause the deterioration of 2,4-DNP degradation and oxidization of VFAs (Wu et al., 2015). Moreover, the 2,4-DNP and UV_{254} degradation rates and effluent AD and OUR in the MBL-MHA were significantly higher than those in the MHA. These results demonstrated that membrane-based bubbleless limited aeration into HA could augment 2,4-DNP degradation and fermentation ability even more than that with bubble micro-aeration into HA.

In addition, the bubble size and aeration intensity not only influence the shear force on sludge and biofilm but also affect the oxygen diffusion efficiency in bulk liquid, sludge, and biofilm inside (Zhao et al., 2021; Li et al., 2004), resulting in different chemical changes (e.g., oxygen and redox potential) and spatial stratification inside sludge and biofilm (Li et al., 2004). The performance of the bioreactor could be affected by microenvironments with different chemical changes and spatial stratifications inside sludge and biofilm (Li et al., 2004). The DO and ORP were measured by microelectrodes to profile the microenvironment of the biofilm in MBL-MHA and the suspended sludge in MHA under different limited aeration conditions (Fig. 3). Three to five biofilm samples from MBL-MHA and suspended sludge samples from MHA were tested. The diameters of the suspended sludge used in this study were determined to be 1.0–1.5 mm. Suspended sludge smaller than 1.0 mm could not be sufficiently stabilized in the batch reactor for accurate microelectrode measurements. Additionally, the 1.0–1.5 mm depths of the biofilm was also determined for accurate microelectrode measurements.

An obviously chemically heterogeneous microstructure was detected in the biofilm and suspended sludge under different limited aeration conditions. For example, for MBL-MHA, the counter-diffusion direction between DO and 2,4-DNP is shown. During phase I, DO inside the biofilm increased from 0 mg/L at the top 0.4 mm of the biofilm to 0.084 ± 0.004 mg/L of membrane-biofilm interface (biofilm depth = 0 μm), and the corresponding ORP values increased from −289.34±0.35 mV to −119.27±0.21 mV. In phase II, DO began to increase from the top 1.3 mm of the biofilm (0 mg/L), and reached 0.28 ± 0.003 mg/L on the membrane-biofilm interface. The corresponding ORP values increased from −276.23 ± 0.26 mV to −75.84 ± 0.18 mV. In phase III, DO increased from 0.12 ± 0.018 mg/L at the top 0.8 mm of the biofilm to 0.39 ± 0.015 mg/L of membrane-biofilm interface, and the corresponding ORP values increased from −104.26 ± 0.18 mV to −67.39 ± 0.21 mV. In phase II, the anaerobic zone was thus limited to the surface layer of the biofilm and bulk liquid, and the remaining biofilm was micro-aerated. In MBL-MHA, a biofilm stratification structure and a significant change in redox status resulting from a single DO gradient in phase II was more obvious than that in phases I and III. For MHA, the co-diffusion direction between DO and 2,4-DNP is shown. For example, during phase I, DO was approximately 0.06±0.017 mg/L in the bulk liquid, and DO inside the suspend sludge was depleted in the top 1.0 mm of the sludge. And the corresponding ORP values decreased from −139.47 ± 0.22 mV to −291.29 ± 0.28 mV. During phase II, DO decreased from 0.23 ± 0.035 mg/L in the bulk liquid to 0 mg/L in the top 1.2 mm in the center of the sludge, according to the decreasing ORP values from −79.49 ± 0.28 mV to −279.29 ± 0.28 mV. In phase III, DO decreased from 0.36 ± 0.032 mg/L in the bulk liquid to 0.21 ± 0.017 mg/L in the center of the sludge, according to the increasing ORP values from −64.24 ± 0.37 mV to −81.25 ± 0.19 mV. In phase II, the micro-aeration zone was thus limited to the surface layer of the suspended sludge and bulk liquid, and the remaining sludge was anaerobic. A more obvious sludge stratification structure resulting from a single DO gradient in phase II was also achieved in MHA compared with that in phases I and III. In addition, because of the opposite diffusion direction of DO between MBL-MHA (counter-diffusion) and MHA (co-diffusion), the redox microenvironments during the degradation of 2,4-DNP were also opposite. For MBL-MHA, because the surface layer of the biofilm and bulk liquid were anaerobic, and the remaining biofilm was micro-aerated, 2,4-DNP could be degraded first by anaerobic degradation on the surface layer of the biofilm and suspended sludge, and then by microaerobic degradation in the biofilm. For MHA, because the surface layer of the suspended sludge and bulk liquid was microaerobic, and the inside of the sludge was anaerobic, 2,4-DNP could be degraded first by microaerobic degradation on the surface layer of the suspended sludge and bulk liquid, and then by anaerobic degradation inside the sludge.

To measure and compare the conversion rates in MBL-MHA and MHA systems, oxygen fluxes at different volumetric locations inside the biofilm and suspended sludge samples were calculated based on our microelectrode measurements (Table 1). Tab. One showed that oxygen fluxes inside the samples from the two micro-aeration HA systems in phase II were higher than those in phases I and III, and under the same limited aeration conditions, those in MBL-MHA were also higher than those in the MHA. These oxygen flux results were lower than those reported for sludge (Li et al., 2004). A possible reason for this difference is that the DO in this study was below 0.5 mg/L, which was much lower than the DO-saturated water used in other studies. The literature reported that under high aeration conditions, the oxygen flux measured by

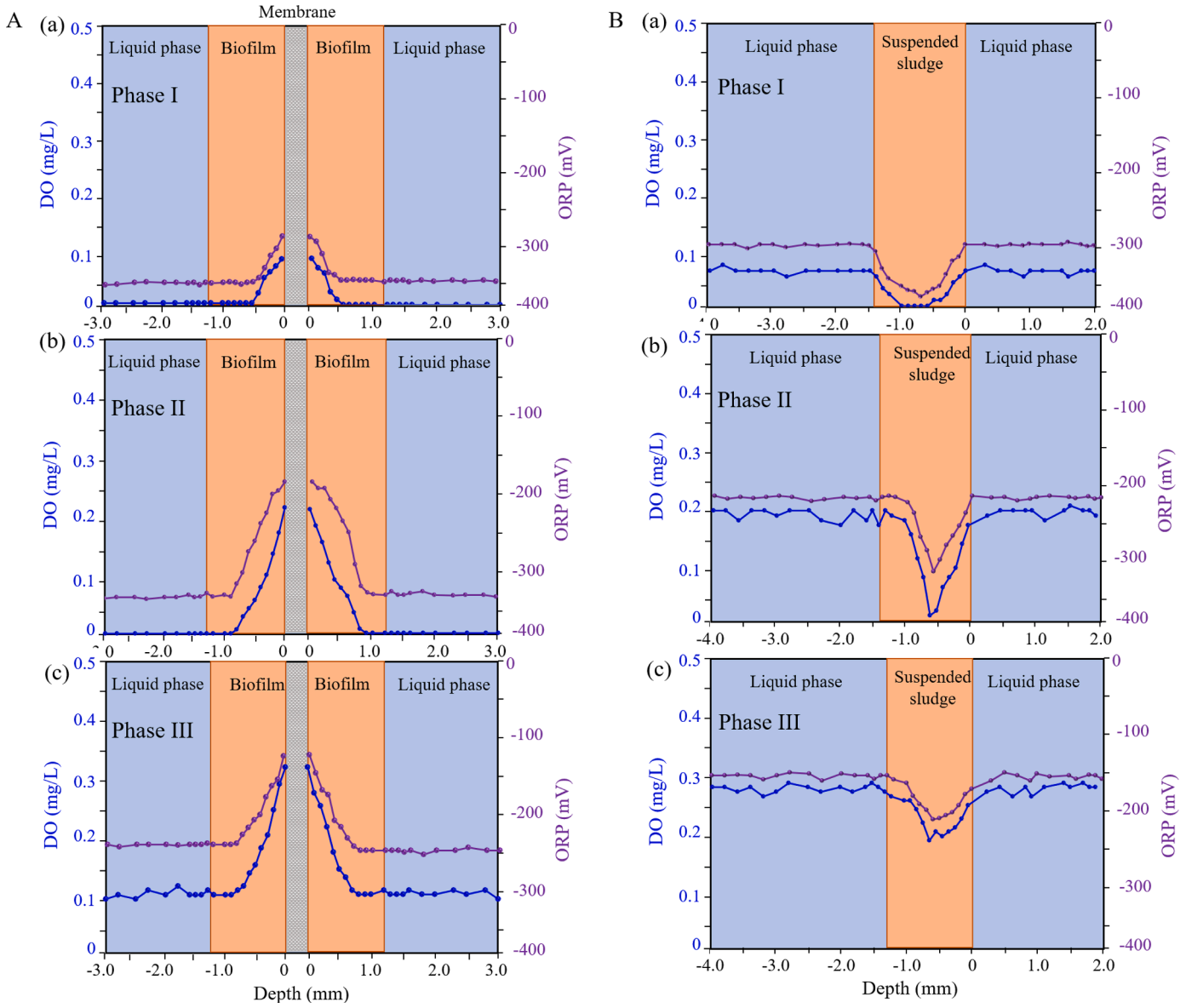


Fig. 3. Microenvironment of sludge in the micro-aeration HA under the different air flow rates (A: biofilm in MBL-MHA; B: suspended sludge in MHA).

Table. 1
Oxygen flux inside biofilm and sludge from MBL-MHA and MHAs.

	MBL-MHA(μmol cm^{-2} h^{-1})	MHA(μmol cm^{-2} h^{-1})
Phase I	0.0175	0.0125
Phase II	0.0372	0.0350
Phase III	0.0325	0.0167

a cathodically polarized oxygen microelectrode was higher than the true oxygen flux (perhaps caused by ignoring the possible influence of convective mass transport and oxygen consumption inside the biofilm), and the standard variation was higher than when a lower DO was used (Li et al., 2004). Some studies have reported that a higher oxygen flux confirmed a higher oxygen utilization capacity and metabolic rates of microorganisms (Schramm et al., 1996; Li et al., 2004). Additionally, diffusion resistance is one of the main factors influencing oxygen flux. And oxygen bubble size is one of the main factors influencing diffusion resistance (Li et al., 2004). Generally, the oxygen transfer coefficient (k_{La} (h^{-1})) has been used to evaluate diffusion resistance (X. Mei et al., 2019). Coefficients of bubble and membrane-based bubbleless micro-aeration in clean water were investigated (Tab. S3). The k_{La} values of the bubbleless limited aeration were 1.23–1.74 h^{-1} under different operational conditions, which was higher than those of the bubble-limited aeration, which were 0.88–1.22 h^{-1}. Wu et al. (2019) reported similar results, that is, k_{La} values increased from 1.33 h^{-1} to 1.42 h^{-1} when the bubble size of aeration decreased from 0.16 μm to 0.10 μm. These results demonstrated that the oxygen utilization capacity and metabolic rates of microorganisms could be enhanced compared with those of bubble-limited aeration, because of the higher oxygen transfer ability of the membrane-based bubbleless limited aeration.

3.2. *Metabolites of 2,4-DNP degradation in MBL-MHA and MHA*

In most cases, even when the parent compound is completely degraded, some intermediate products remain in the water environment (Li et al., 2021). Tang et al. (2018) investigated the degradation products of 2,4-DNP under anoxic conditions and found that 2,4-DAP, 2-A-4-NP, 2-N-4-AP, phenol, and small-molecule acids were produced under anoxic conditions. In this study, six main intermediate products were detected by LC-MS in MBL-MHA: 2,4-DAP, 2-A-4-NP, 2-N-4-AP, 2-NP, 4-NP, phenol, and benzoic acid (Fig. S3). And 4-Nitrocatechol,

2-Nitrohydroquinone, and benzenethiol were mainly detected by LC-MS in MHA (Fig. S3). These intermediate products were different between MBL-MHA and MHA, indicating that the degradation pathways of 2, 4-DNP might be different.

To analyze the denitration characteristics of 2,4-DNP in the MBL-MHA and MHA systems, this study investigated nitrogen transformations within a given HRT by using the batch reaction mode under phase II (Fig. S4). The NH_4^+-N concentration increased rapidly and then decreased in MBL-MHA, and it remained relatively stable close to zero in the MHA. For MBL-MHA, the nitro groups on the benzene ring were first reduced to amino groups, generating the corresponding aminophenols, such as 2,4-DAP, 2-A-4-NP, and 2-N-4-AP. In addition, the NH_4^+-N concentration decreased rapidly, and the NO_2^--N concentration increased rapidly and then decreased, indicating that nitrosification and denitrification might occur simultaneously in MBL-MHA. NH_4^+-N could be conversed to NO_2^--N at DO of 0.2–0.4 mg/L (Wang et al., 2009). For MHA, the NO_2^--N concentration increased more rapidly than in MBL-MHA, probably because of the accumulation of NO_2^--N caused by the shedding of nitro groups. In addition, the NO_2^--N concentration generally decreased, indicating that denitrification might have occurred in MHA. Based on the properties of 2,4-DNP and the products detected from LC-MS, the possible pathways of 2,4-DNP degradation in MBL-MHA were nitroreduction, deamination from the benzene ring, and simultaneous nitrosification and denitrification, and the possible pathways in MHA were hydroxylation, nitro group removal from the benzene ring, and denitrification (Fig. 4). These results were due to the counter-diffusion in MBL-MHA and co-diffusion in MHA, which might lead to different degradation pathways of 2,4-DNP. In MBL-MHA, 2, 4-DNP could be transferred first to the corresponding aminophenol, such as 2-A-4-NP, 2-N-4-AP, and 2,4-DAP in the suspended sludge and the surface layer of the biofilm through the nitroreduction process under anaerobic conditions (Jiang et al., 2016). Next, the corresponding aminophenol could be further degraded to phenol through the deamination process (Tang et al., 2018). In addition, phenol could be degraded to benzoic acid and then to acetate under micro-aeration in the biofilm. In MHA, hydroxylation and denitration of 2,4-DNP could occur on the outer layer of the suspended sludge under micro-aeration, and nitrite and benzenethiol were produced. Subsequently, nitrite can be removed through the denitrification process.

3.3. Variations of EPS

Fig. 5(A) and 5(B) show the variations in the LB-EPS and TB-EPS contents (polysaccharide, protein, and nucleic acid) of MBL-MHA and MHA under different limited aeration conditions and anaerobic HA. Fig. 5(A) and 5(B) show that protein content was more dominant than polysaccharide and nucleic acid contents in EPS production in the three systems. The protein content in phase II was higher than that in phases I and III, and that in biofilms in MBL-MHA was higher than that in suspended sludge in MBL-MHA and MHA. The protein of EPS is mainly related to extracellular enzymes or structural proteins (Tian et al., 2021). Extracellular enzymes could be important for capturing or adhering to water-soluble and water-insoluble organic compounds in the matrix, providing nutrients for the microbe growth and promoting the transformation of organic matter. Structural proteins (non-enzymatic proteins) are the main components of EPS proteins, such as cell-surface binding and extracellular carbohydrate-binding proteins. Appropriate, limited aeration into HA could promote various

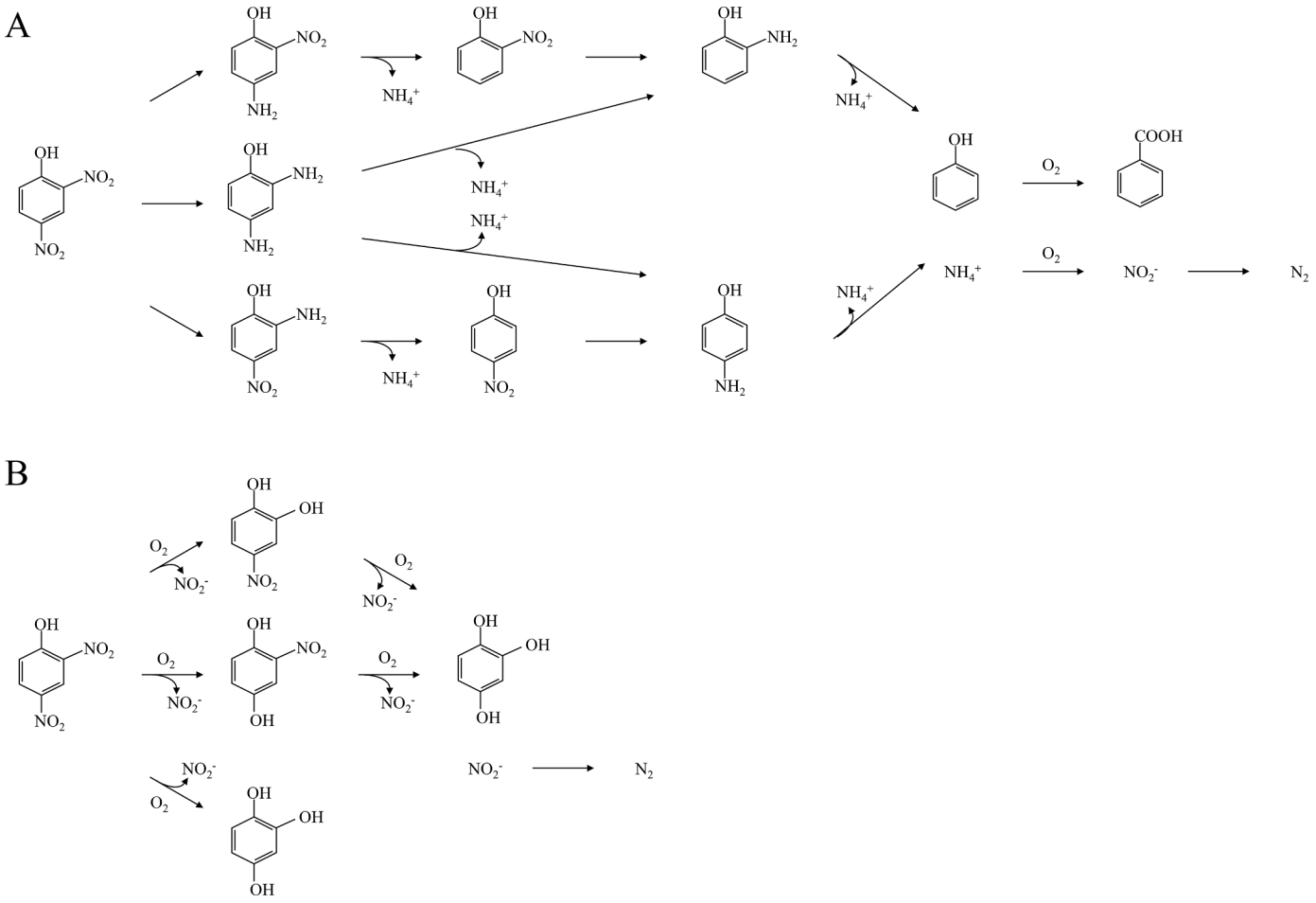


Fig. 4. Proposed degradation pathways of 2,4-DNP in MBL-MHA (A) and MHA (B).

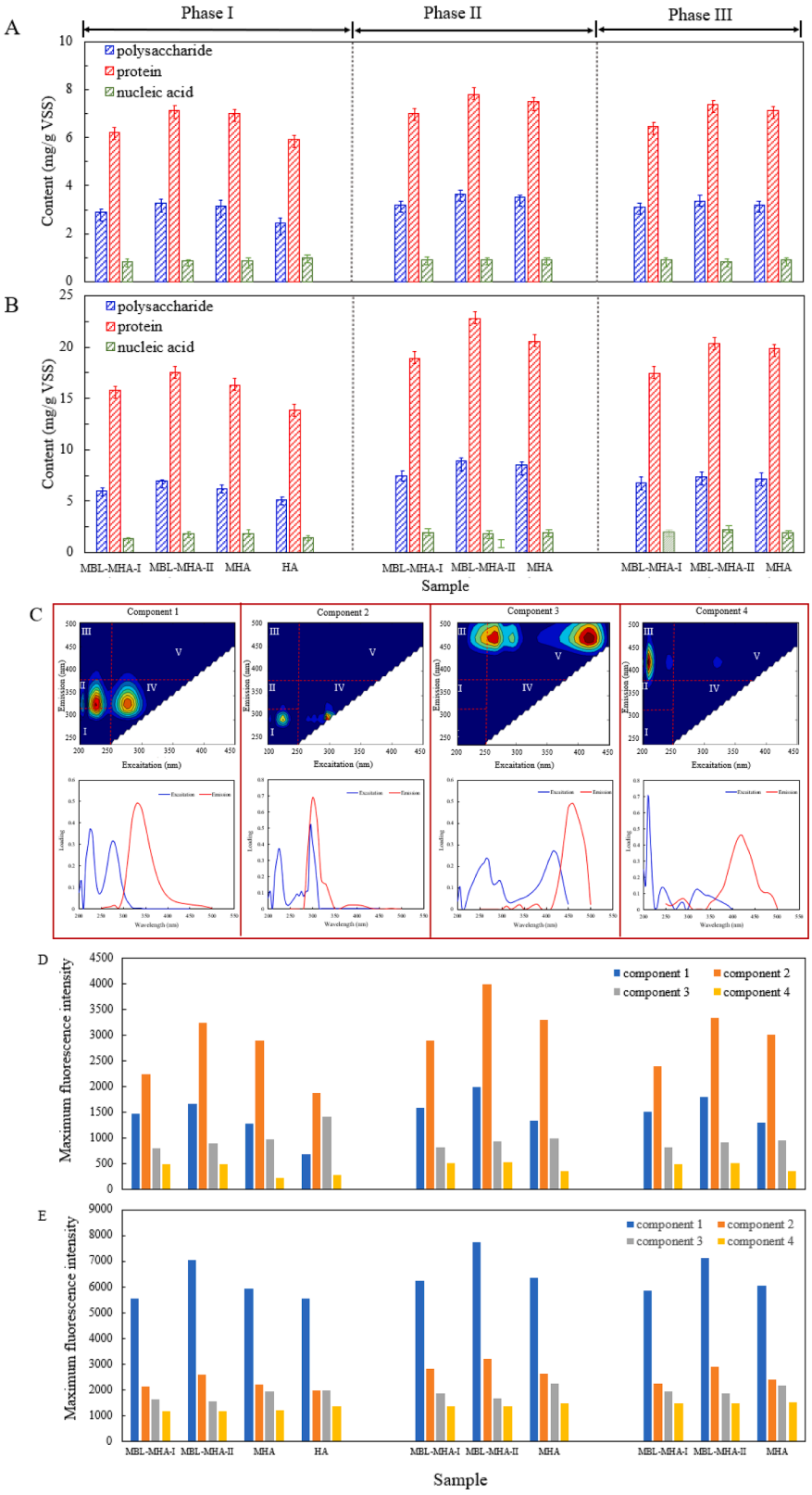


Fig. 5. Variations of LB-EPS content (A) and TB-EPS content (B); 3D-EEM spectra of three fluorescence components with PARAFAC analysis (C) and maximum fluorescence intensities (Fmax) of three PARAFAC-derived components of LB-EPS (D) and TB-EPS (E) at different air rates.

extracellular enzyme activities (Ruan et al., 2019), which could be one of the main reasons for the increased protein content of EPS in phase II. Additionally, Zhang et al. (2015) reported that compared with that in suspended sludge, EPS protein in biofilms also contained more structural proteins to promote cell adhesion. The EPS protein could also affect the morphologies of sludge and biofilm (Chen et al., 2021). The morphologies of suspended sludge of MBL-MHA, biofilm on the membrane surface of MBL-MHA, and suspended sludges of MHA and anaerobic HA

during stable operation in phase II were observed through SEM in Fig. S7. From Fig. S7, the rod-shaped bacteria and spheroidal bacteria were dominant under micro-aeration, and the suspended sludge under micro-aeration was more dense than that in HA. Meanwhile, a certain thickness of biofilm was uniformly attached to the membrane surface, and the structure of biofilm was more dense than the suspended sludge.

Moreover, the protein content of TB-EPS was higher than that of LB-EPS in the four samples. Similar results have been reported by Ye et al. (2011) and Zhang et al. (2014). These results might occur because the compositions and properties of LB-EPS are unstable and easily influenced by external conditions. By contrast, the stable composition and function of TB-EPS are less affected by the environment (Ye et al., 2011). Studies have reported that TB-EPS have strong biosorption ability, and their functional groups play a significant role in the biosorption of suspended sludge and biofilms (Gao et al., 2011; Zhang et al., 2014).

The 3D-EEM spectroscopy and PARAFAC methods have been proved to be appropriate methods for EPS analysis (Sheng and Yu, 2006); thus, they were also used in this study for EPS analysis. The PARAFAC models of TB-EPS and LB-EPS in the four samples with different micro-aeration conditions revealed that the fluorescence could be characterized by four components (Fig. 5(C)). Component 1 resembled the tryptophan-like protein (Ex/Em 225/350 nm) and soluble microbial byproduct-like matter (Ex/Em 275/350 nm) (Chen et al., 2003). Component 2 mainly showed tyrosine-like fluorescence (Ex/Em 220/300 nm) (Chen et al., 2003; Guo et al., 2012). Components 1 and 2 have been ascribed to protein-like fluorophores (Baghoth et al., 2011). Component 3 exhibited F_{max} at Ex/Em wavelengths of 265/475 nm and 425/475 nm, which are associated with humic acid-like fluorescence (Chen et al., 2003). Component 4 mainly showed fulvic acid-like fluorescence (Ex/Em 220/300 nm) (Chen et al., 2003).

PARAFAC analysis also provided quantitative information based on the F_{max} values of the four components in the LB-EPS and TB-EPS samples under varying micro-aeration conditions (Fig. 5(D), (E)). The protein-like substances (components 1 and 2) were major constituents in the LB-EPS and TB-EPS samples, and those in the TB-EPS were higher than those in the LB-EPS. The F_{max} of components 1 and 2 in TB-EPS in phase II was higher than that in phases I and III, and that in the biofilm in MBL-MHA was higher than that in suspended sludge in MBL-MHA and MHA, which indicated that there were more abundant tyrosine-like and tryptophan-like protein substances in TB-EPS in phase II and in the biofilm in MBL-MHA. These results were consistent with the protein content in the TB-EPS samples during phases I–III. The increased F_{max} of protein-like substances in TB-EPS in phase II demonstrated that it played a certain role in organic matter removal and transformation in the biosorption process (Song et al., 2019). Aquino et al. (2004) investigated the interaction between protein-like substances in EPS and chloroform, indicating that the interaction increased as the amount of protein-like substances increased.

To further investigate the changes in the chemical structures and functionalities of EPS in MBL-MHA, MHA, and anaerobic HA, UV–*vis* and FT-IR qualitative spectra of LB-EPS and TB-EPS from biofilm in the MBL-MHA and suspended sludge in the MBL-MHA and MHA were used in phase II (Fig. S5 and S6). Fig. S5 shows that there was a weak absorption peak at 275 nm in LB-EPS and TB-EPS in the three systems. The UV–*vis* absorption peaks of aromatic proteins in EPS generally range from 245 nm to 290 nm, which belong to the *n*-Π* and Π-Π* electron transfer of the C = O and C = C groups from tryptophan and tyrosine (He et al., 2020). The absorbance of MBL-MHA-II was lower than that of MHA and anaerobic HA, and the protein content in TB-EPS of MBL-MHA-II was higher than that of MHA and anaerobic HA, indicating that stronger interactions among TB-EPS, 2,4-DNP, and its transformation products might occur in the biofilm (He et al., 2020).

A general belief is that the 1800–600 cm^{-1} region of the FT-IR spectra can provide information on the compositions, chemical structures, and functionalities of EPS components (Badireddy et al., 2010; Zhang et al., 2015; Wang et al., 2017). Fig. S6 shows the FT-IR qualitative spectra of LB-EPS and TB-EPS, which mainly included six predominant peaks associated with several characteristic functional groups. In general, the FT-IR profiles for MBL-MHA, MHA, and HA were similar, indicating that the main functional groups were similar for these EPSs. The peaks at 1650 cm^{-1}, 1525 cm^{-1}, and 1425 cm^{-1} were assigned to the C = O stretching vibration of the amide I, N–H deformation vibration of amide II and the C–N stretching vibration of amide III, respectively (Badireddy et al., 2010; Zhang et al., 2015). The band at 1100 cm^{-1} was related to the C–O–C and C–H stretching vibrations of the polysaccharide structures (Wang et al., 2012). The band at 1100 cm^{-1} may also be associated with the $P = O$ and C–P–O vibrations (Badireddy et al., 2010; Fang et al., 2012). These results indicated that extracellular proteins could bind or mix carbohydrates and phosphorylated macromolecules. The ranges of 600–900 cm^{-1} belonged to the fingerprint region, which mainly reflects the presence of carbohydrates and nucleic acids (Badireddy et al., 2010; Yin et al., 2015). The FT-IR spectra showed that the EPS contained proteins, polysaccharides, nucleic acids, and other substances, which was consistent with the composition of EPS. According to the literature (Zhang et al., 2019), extracellular enzymes with oxidoreductase activity in EPS in HA generally act on the CH—OH/CH—CH/CH/CH_2 group of donors, and those with hydrolase activity generally act on C—N. In addition, redox-active oxygen-containing functional groups, such as C = O, can facilitate microbial reduction of organic matter (Wu et al., 2017). Ren et al. (2020) revealed that surface oxygen-containing functional groups of hydrochar could mediate direct interspecies electron transfer in an anaerobic digestion system. Xiao et al. (2017) reported that redox proteins in EPS could be the transient media of interspecies interactions through its function of interspecific electron transfer to promote the removal of organic matter. These peaks of redox-active functional groups in TB-EPS in FT-IR were stronger than those in LB-EPS, and those of biofilm in MBL-MHA was stronger than that of suspended sludge in MHA and anaerobic HA, indicating a greater abundance of redox proteins with oxidoreductase activity in the biofilm of the MBL-MHA.

3.4. Microbial community analysis

Illumina MiSeq high-throughput sequencing was used to reveal the microbial community structure of the different sludge samples. The community richness and alpha diversity indices of the samples are shown in Table 2. The coverage of each sample was 0.999, indicating that the sequence analysis represented an almost complete microbial population. The ACE and Chao indices indicate microbial community richness, and the Shannon and Simpson indices represent microbial community diversity (Jena et al., 2016).

The observed ACE and Chao indices in the samples from MBL-MHA and MHA were higher than those from anaerobic HA, indicating that the microbial community became relatively more diverse under micro-aeration conditions. Additionally, Shannon indices in the samples from MBL-MHA and MHA were higher than those from anaerobic HA, and Simpson indices in the samples from MBL-MHA and MHA were closer to 0 than those from anaerobic HA, indicating that the microbial community was richer under micro-aeration conditions. These results indicate that suitable limited aeration in HA could increase microbial

Table. 2
Community richness and diversity indices of different samples.

Sample	OTUs	Ace	Chao	Shannon	Simpson	Coverage
MBL-MHA-I	564	599	590	4.02	0.050	0.999
MBL-MHA-II	607	621	603	4.32	0.042	0.999
MHA	586	609	593	4.11	0.048	0.999
Anaerobic HA	528	567	587	3.87	0.059	0.999

Note: the suspended sludges of MBL-MHA (MBL-MHA-I), the biofilm of MBL-MHA (MBL-MHA-II), the suspended sludges of MHA (MHA), and the suspended sludges of anaerobic HA (Anaerobic HA).

community richness and diversity. This result is similar to that reported (Nguyen et al., 2019; Krayzelova et al., 2015). Notably, the samples from MBL-MHA-II showed a higher diversity (Shannon index of 4.32; Simpson index of 0.042) and richness (ACE index of 621; Chao index of 603) than those from MHA (Shannon index of 4.11; Simpson index of 0.048; ACE index of 609; Chao index of 593), indicating that the microbial community richness and diversity became relatively richer under the membrane-based bubbleless micro-aeration in HA than that under the bubble micro-aeration in HA. High biodiversity and richness can increase ecological stability, which would result in a high capacity to resist environmental stress (Zhang et al., 2013). The Venn diagram of these samples (Fig. 5(C)) shows that the OTUs shared by MBL-MHA-I, MBL-MHA-II, MHA, and anaerobic HA accounted for 23% of the total OTUs, and the number of unique OTUs was 46, 113, 93, and 62, respectively. According to the results of the biofilm microenvironment (Fig. 3), the microbial community structures from the

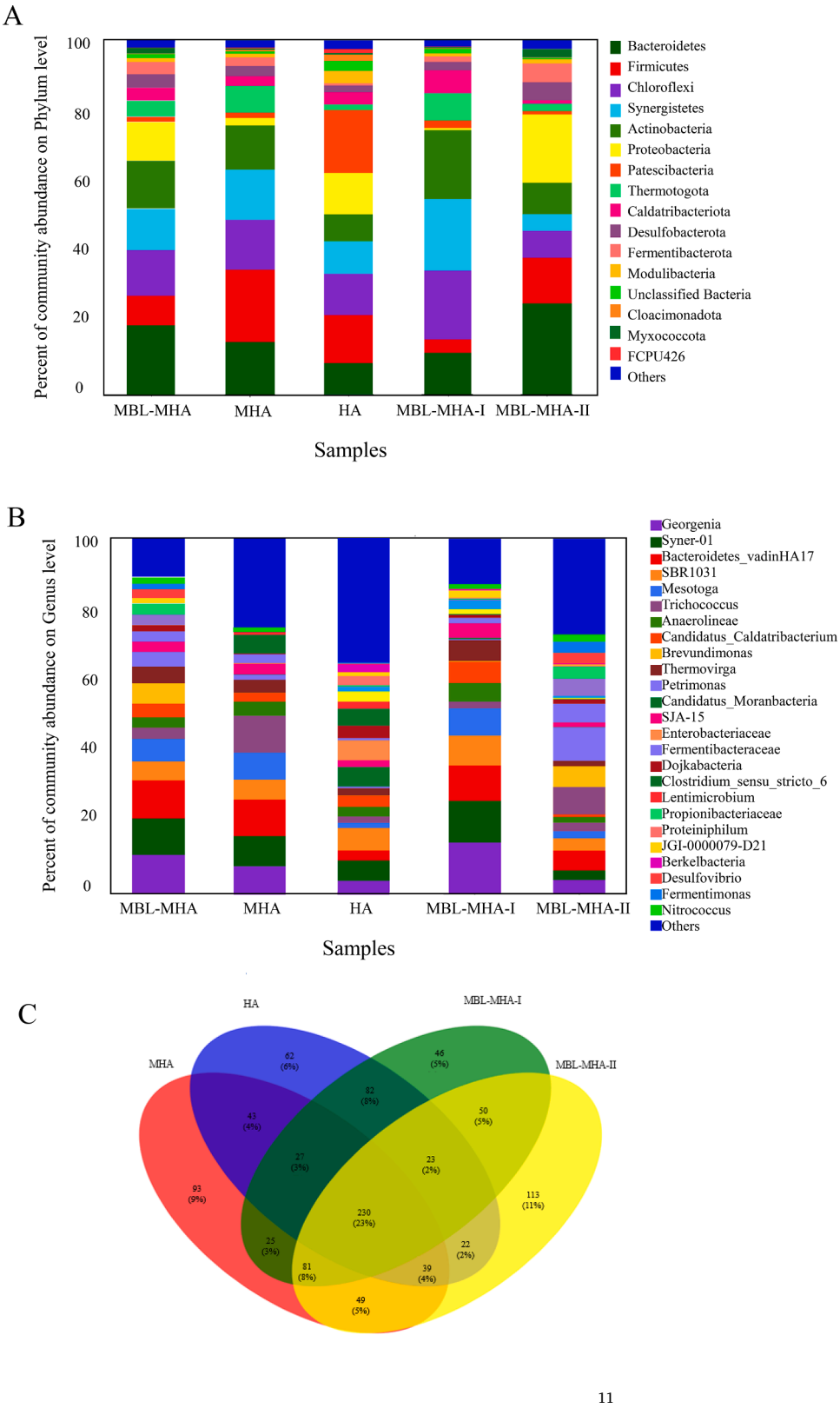


Fig. 6. Relative abundance of the dominant bacterial community at (A) phylum level and (B) genus level; (C) Venn diagram of samples MBL-MHA-I, MBL-MHA-II, MHA, and HA
Note: MBL-MHA: total of suspended sludge and biofilm on the membrane surface, MBL-MHA-I: suspended sludge of MBL-MHA, MBL-MHA-II: biofilm on the membrane surface of MBL-MHA, MHA: suspended sludges of MHA, HA: suspended sludges of HA.

membrane-biofilm interface (biofilm depth = 0 μm) to bulk liquid were divided into more obvious micro-aeration and anaerobic zones, which resulted in increased greater biodiversity and explain why the OTU number of MBL-MHA-II was higher than that of MHA (Ma et al., 2017; X. Mei et al., 2019). The higher synergistic effects of various microorganisms in MBL-MHA resulted in higher efficient 2,4-DNP degradation.

The diversity of microorganisms at the phylum and genus levels in the three reactors is shown in Fig. 6. The Illumina results indicated that the bacterial community was mainly composed of 37 phyla, except for unclassified bacteria in MBL-MHA, 35 phyla in MHA, and 32 phyla in anaerobic HA. As shown in Fig. 6A, the relative microbial community abundances were identified at the phylum level, and Bacteroidetes, Firmicutes, Chloroflexi, Synergistetes, Actinobacteria, and Proteobacteria were dominant in these five samples, with minor phyla accounting for a small fraction.

The main differences between the micro-aeration HA and the anaerobic HA systems were attributed to the distribution of phyla Bacteroidetes, Firmicutes, Chloroflexi, Synergistetes, and Actinobacteria. Relative abundances of the phyla Bacteroidetes, Firmicutes, Chloroflexi, Synergistetes, and Actinobacteria were 16.15%, 18.4%, 16.81%, 12.46%, and 14.84% in MBL-MHA, and 14.74%, 20.45%, 14.12%, 14.17%, and 12.46% in MHA, respectively, which were significantly higher than those in anaerobic HA, that is, 8.80%, 13.56%, 11.69%, 9.17%, and 7.62%, respectively, indicating that limited aeration could help selectively enrich specific species. However, relative abundances of the phyla Proteobacteria and Patescibacteria were 5.13% and 2.14% in MBL-MHA and 2.10% and 1.53% in MHA, respectively, which was lower than that in anaerobic HA (11.67% and 17.80%, respectively). Additionally, relative abundances of these phyla in the MBL-MHA were also obviously higher than those in MHA, indicating that MBL-MHA was more favorable for selective enrichment of some specific species than MHA.

Additionally, the relative bacterial community abundances of the suspended sludge (MBL-MHA-I) and biofilm (MBL-MHA-II) in MBL-MHA were also identified at the phylum level (Fig. 6A). The relative abundances of Bacteroidetes, Firmicutes, and Proteobacteria in the biofilm were 25.62%, 12.88%, and 19.30%, respectively, and significantly higher than those in the suspended sludge (11.68%, 3.77%, and 0.62%, respectively). Nevertheless, the relative abundances of Chloroflexi, Synergistetes, and Actinobacteria in the suspended sludge (19.45%, 20.21%, and 19.43%) were obviously higher than those of the biofilm (7.67%, 4.66%, and 8.84%, respectively). These results indicated that spatial distribution of the specific bacteria could be achieved using membrane-based bubbleless micro-aeration. A similar spatial distribution structure has been reported (Wang et al., 2009; Kinh et al., 2017; Ma et al., 2017). For example, Kinh et al. (2017) found that nitrifying bacteria and denitrifying bacteria were gradually distributed from the membrane surface to the bulk liquid, which realized simultaneous nitrification and denitrification. These results could mainly be attributed to the following: a single DO gradient structure within the biofilm from the membrane-biofilm interface to the bulk liquid caused by counter-diffusion was clearly presented, showing spatial variations of some specific species (Kinh et al., 2017; Ma et al., 2017).

The next objective was to gain insight into the microbial community structure in the three sludge samples; thus, the sequences from the dominant bacterial communities were analyzed at the genus level as follows (Fig. 6B). Compared with the phylum level classifications, the results of the genus level illustrated a greater diversity in the microbial communities in MBL-MHA, MHA, and anaerobic HA samples, which were composed of 334, 318, and 262 genera, respectively. As shown in Fig. 6B, differences were observed despite most of the genera being found in the five samples. *Syner-01* (5.64%), *Georgenia* (4.14%), and *SBR1031* (6.24%) were dominant in anaerobic HA. *Syner-01,* belonging to the phylum Synergistetes, includes hydrolytic bacteria with the ability to produce acetate through glucose fermentation (Si et al., 2016). *Georgenia,* phylum Actinobacteria, is associated with the production of abundant hydrolytic enzymes that utilize organic matter such as glucose and is related to butyric acid production (Liu et al., 2016; Chen et al., 2020a, b, c; Nelson et al., 2011). *SBR1031*, phylum Chloroflexi, has been widely reported to utilize various substrates, such as organic acids, and have strong resistance to toxic organic pollutants (Chen et al., 2020a, b, c; Nakasaki et al., 2019). Studies have demonstrated that Actinobacteria and Chloroflexi are important 2,4-DNP-degrading bacteria in anaerobic conditions (Kimura et al., 2003; Arnett et al., 2009; Yang et al., 2015), and under anaerobic conditions, these bacteria can degrade 2,4-DNP into corresponding aminophenols, including 2,4-DAP, 2-A-4-NP, and 2-N-4-AP (Kimura et al., 2003; Tang et al., 2018). Additionally, under the micro-aeration condition, these facultative bacteria could, by using molecular oxygen, further accelerate the decomposition of these aminophenols, resulting in the NH_4^+-N release (An et al., 2020).

In both micro-aeration HA systems, *Bacteroidetes_vadinHA17* (7.55% of MBL-MHA; 10.18% of MHA), *Mesotoga* (4.55% of MBL-MHA, 7.54% of MHA), and *Trichococcus* (2.17% of MBL-MHA; 10.28% of MHA) had higher relative abundances than those in anaerobic HA (2.07%, 1.46%, and 1.78%). Moreover, *Lentimicrobium* (2.78% of MBL-MHA; 0.78% of MHA), *Petrimonas* (5.38% of MBL-MHA, 1.44% of MHA), and *Fermentibacteraceae* (3.61% of MBL-MHA; 2.46% of MHA) were remarkably enriched in the two limited aeration HA systems; nevertheless, they showed the least relative abundance in anaerobic HA (0.15%, 0.47%, and 0.62%, respectively). *Fermentibacteraceae* and *Trichococcus* within Firmicutes can convert various organics to VFAs by secreting various extracellular enzymes and subsequently to acetate, H_2, and CO_2 (Zhang et al., 2018; Owusu-Agyeman et al., 2021). Specifically, the ability to tolerate high organic loading and toxic substances has also been reported (Jia et al., 2019; Yang et al., 2016; Ma et al., 2020). *Lentimicrobium* and *Bacteroidetes_vadinHA17* are highly related to the treatment of nitrate wastewater and denitrification processes (Niu et al., 2021; H. Wang et al., 2020; Mielcarek et al., 2016). Notably, *Petrimonas* can metabolize amino acids to ethanol by transferring electrons to the limited molecular oxygen, resulting from the promotion of interspecific electron transfer in the HA system (Wang et al., 2021; Zhao et al., 2021). Botheju et al. (2009) reported that trace oxygen in an anaerobic bioreactor could act as a terminal electron acceptor to accelerate the hydrolysis of long-chain organic compounds under facultative bacteria, improving the electron transfer activity of microbial metabolism. The relative abundance of this genus in MBL-MHA was significantly higher than that in MHA, indicating that the addition of micro-aeration in HA could stimulate electron transfer activity of microbial metabolism, especially membrane-based bubbleless limited aeration. Moreover, compared with that in MHA and anaerobic HA, the more differences were attributed to the distributions of the genera *Brevundimonas* (7.53%) and *Nitrosomonas* (2.95%) in MBL-MHA. *Brevundimonas*, belonging to Proteobacteria, could enhance the decomposition of nitrogen-containing organic matter, such as aminophenols, by using limited aeration, resulting in the release of NH_4^+-N (He et al., 2021). And *Nitrosomonas* within Proteobacteria oxidized NH_4^+-N to NO_2^--N by using free oxygen under a DO below 0.5 mg/L (H. Wang et al., 2020). In MBL-MHA, relative abundances of *Brevundimonas* (13.44%), *Nitrosomonas* (2.08%), *Lentimicrobium* (4.85%), and *Petrimonas* (9.46%) from the biofilm were remarkably higher than those of the suspended sludge, that is, 0.01%, 0.01%, 0.21%, and 0.30%, respectively, which might attribute to the fact that these genes were more attuned to micro-aeration condition. The *Syner-01* (11.75%), *Georgenia* (14.32%), and *SBR1031* (8.46%) of the suspended sludge were richer than those of the biofilm (2.71%, 3.76%, and 3.41%, respectively).

According to the potential functions of these domain bacteria in the two micro-aeration HA systems, these dominant bacteria can be divided into four groups: reduction-associated bacteria, oxidation-associated bacteria, interspecific electron transfer-associated bacteria, and fermentative bacteria. In MHA, the oxidation-associated bacteria, such as *Syner-01, Georgenia,* and *SBR1031*, could play a major role in denitration from nitrophenol. Reduction-associated bacteria, such as

Lentimicrobium and *Bacteroidetes_vadinHA17*, could play a major role in denitrification. The interspecific electron transfer-associated bacteria were mainly represented by *Petrimonas*, which could improve the activity of microbial metabolism in HA. Fermentative bacteria, such as *Fermentibacteraceae* and *Trichococcus*, mainly convert the degradation intermediates of nitrophenol to VFAs. Unlike the MHA, the reduction and oxidation-associated bacteria in the MBL-MHA exhibited different functions in degrading 2,4-DNP. Specifically, reduction-associated bacteria could play a major role in ammoniation for reducing 2,4-DNP into corresponding aminophenols and denitrification, such as *Syner-01, Georgenia, SBR1031, Lentimicrobium*, and *Bacteroidetes_vadinHA17*. These bacteria were mainly enriched in suspended sludge. The oxidation-associated bacteria could play a major role in the deamination of aminophenols and nitrosation, such as *Brevundimonas* and *Nitrosomonas*. These bacteria were mainly enriched in biofilms. The fermentative bacteria could mainly convert the degradation intermediates of 2,4-DNP to VFAs, such as *Fermentibacteraceae* and *Trichococcus*. The interspecific electron transfer-associated bacteria were mainly represented by *Petrimonas*, which could improve the activity of microbial metabolism in the HA system. These species were mainly enriched in the biofilm.

As indicated, improved biodiversity, enrichment of reduction-related and oxidation-related species within syntrophic interactions, improvement in interspecific electron transfer-related species, and fermentative species might play a major role in enhancing 2,4-DNP degradation in micro-aeration HA. Moreover, these functional bacteria could be more enriched using membrane-based bubbleless micro-aeration because of a more obvious spatial distribution of the specific bacteria in the biofilm.

3.5. Label-free quantitative proteomic profiling

Proteins are the products and functional units of the genome. However, the genome is not representative of protein levels, because of the complex systems governing when and how much of a given protein should be present (Zhang et al., 2019). Hence, further investigations on the metaproteome and its relationship with the metagenome in MBL-MHA, MHA, and anaerobic HA were prompted by label-free proteomic profiling. Four assays in phase II (i.e., MBL-MHA-I, MBL-MHA-II, MHA, and anaerobic HA) were analyzed.

A total of 456,768 secondary spectra were detected, 22,801 peptides were matched, and 22,801 peptide sequences and 7362 unique protein groups were identified in the four assays. The number of proteins identified in each MBL-MHA-I, MHA-II, MHA, and anaerobic HA was 5157, 5397, 5232, and 5081, respectively. Quality assessment of the identified proteins revealed that the values/data had good repeatability, high quality, and credibility. In this study, proteins were considered to be significantly differentially expressed (Table 3). Compared with the proteins expressed in the anaerobic HA assay, the number of upregulated proteins in samples MBL-MHA-I, MBL-MHA-II, and MHA amounted to 1508, 1574, and 1504, respectively. These results indicate that the addition of limited aeration in HA for the degradation of 2,4-DNP significantly affected the differential expression of functional proteins.

Table. 3
Number of different expressed proteins in grouped samples (FC > 1.2 & $p < 0.05$ or FC < 0.83 & $p < 0.05$).

Sample	Number		
	Up-regulated proteins	Down-regulated proteins	Non-significant differential proteins
MBL-MHA-I (MBL-MHA-I vs Anaerobic HA)	1508	1258	1957
MBL-MHA-II (MBL-MHA-II vs Anaerobic HA)	1574	2138	314
MHA (MHA vs Anaerobic HA)	1504	1659	1518

Note: FC-fold change; p-p-value, which represented the statistically difference in protein expression.

3.5.1. Gene ontology (GO) functional annotation of upregulated proteins

Based on the GO functional annotation of upregulated proteins, as shown in Fig. 7A, the upregulated proteins of MBL-MHA-I, MBL-MHA-II, and MHA were associated with biological processes (BP), cell components (CC), and molecular functions (MF). Compared with those in the assay anaerobic HA, the upregulated proteins in the assays MBL-MHA-I, MBL-MHA-II, and MHA were classified into 24 functional groups (12 BP, 2 CC, 10 MF). One protein may be involved in different BP, and different proteins may be involved in the same process. In the BP analysis, against the assay anaerobic HA, the majority of the upregulated proteins in the limited aeration assays were significantly assigned to the categories "metabolic process," "interspecies interaction between organisms," and "cellular process" ($p \leq 0.05$, and protein number ≥ 500). Interspecies interactions can affect bacterial structure, cell abundance, and stress responses. The enhanced interspecies interactions could display stronger stress tolerance, rendering them substantially more robust for the treatment of toxic organic matter (K.W.K. Lee et al., 2014). The RichFactor of "interspecies interaction between organisms" in the assay MBL-MHA-II was 0.99, and that in the assay MHA and MBL-MHA-I was 0.83 and 0.35, respectively, indicating that compared with bubble micro-aeration in HA, the proteins with interspecies interactions could be more enhanced in membrane-based bubbleless micro-aeration in HA, and this enhancement was primarily enriched in the biofilm. According to the literature, an obvious DO gradient from the membrane biofilm to the bulk liquid, and a rich microbial community structure in the biofilm could show an enhanced interspecies interaction (X. Mei et al., 2019). In addition, the redox protein in EPS could be the transient medium of interspecies interactions via its function of interspecific electron transfer (Xiao et al., 2017). In this study, the more abundant redox protein in TB-EPS in the biofilm on the membrane of MBL-MHA might promote this process more than MHA. These results implied one of the underlying reasons that using the membrane-based bubbleless micro-aeration into HA could enhance the degradation of 2,4-DNP even more than bubble micro-aeration. In the CC analysis, most of the upregulated proteins in the micro-aeration HA assays belonged to "Protein-containing complex, " followed by "Cellular anatomical entity".

In the MF process, most of the upregulated proteins in micro-aeration HA were related to "binding" and "catalytic activity" functions. The category "catalytic activity" included oxidoreductase activity acting on the CH—OH/aldehyde/oxo/NAD(P)H/$CHNH_2$/CH—CH/CH/CH_2 group of donors, and hydrolase activity acting on acid anhydrides/carbon-nitrogen/ester/glycosyl/acid halide and some transmembrane transporter activities (Zhang et al., 2019). The RichFactor of this activity function in the MBL-MHA-II assay was 0.94, which was higher than that of MHA (0.62) and MBL-MHA-I (0.55). These findings suggest that micro-aeration HA could enhance the expression of proteins related to oxidoreductase activities, ring cleavage, and intermediate transmembrane processes for 2,4-DNP biodegradation better than anaerobic HA, especially MBL-MHA.

3.5.2. Kyoto encyclopedia of genes and genome (KEGG) pathway annotation of upregulated proteins

To further verify the metabolic and signaling pathways for the upregulated proteins in MBL-MHA and MHA compared with those of anaerobic HA, the identified proteins were annotated with KEGG. KEGG pathway enrichment analysis further showed that limited aeration in HA-enriched proteins was involved in various pathways, such as nitrogen metabolism, carbon metabolism, glycolysis/gluconeogenesis, peroxisome, oxidative phosphorylation, and propanoate metabolism, which are directly related to the efficiency of hydrolysis, fermentation,

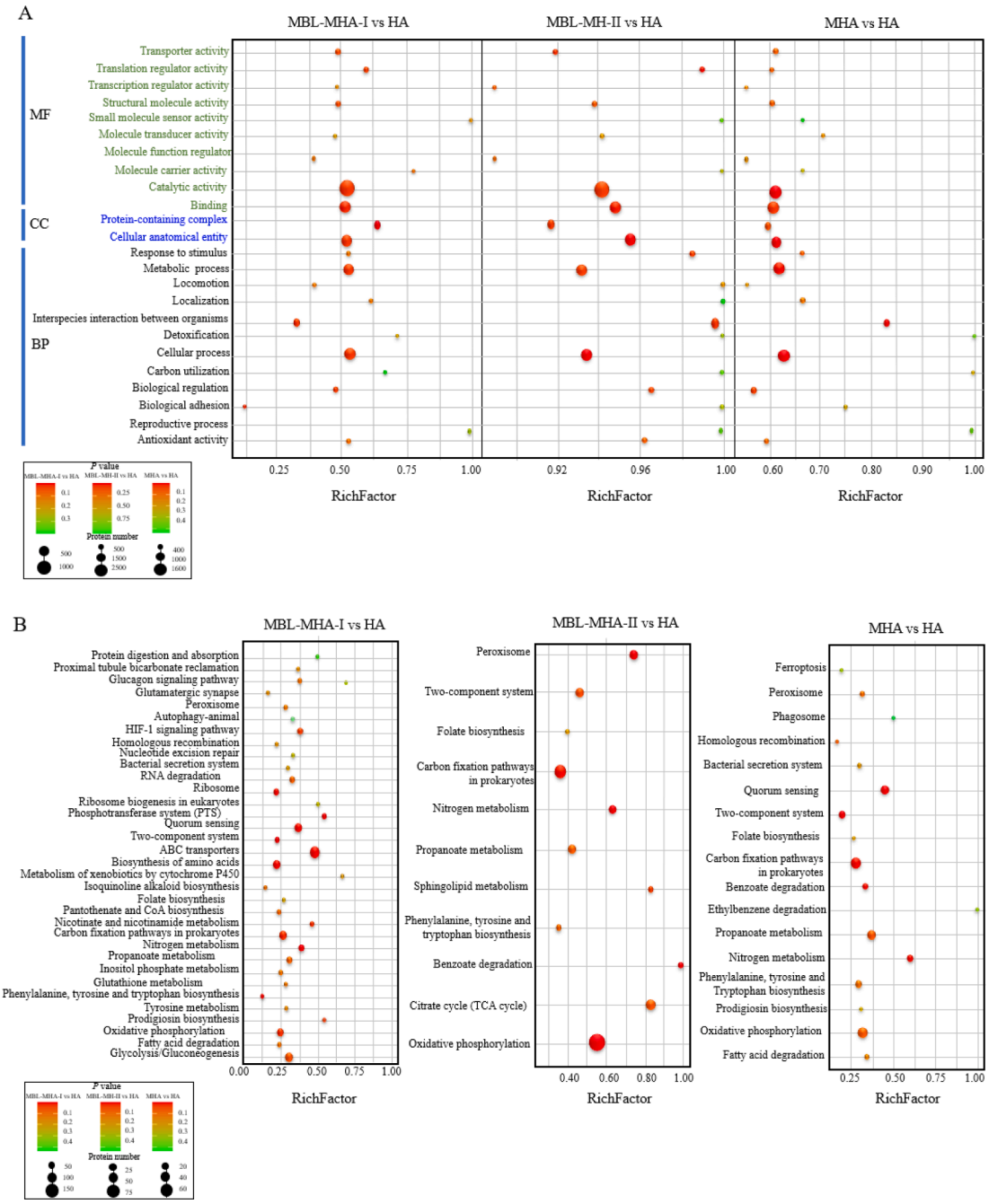


Fig. 7. The GO (A) and KEGG (B) functional annotation of different up-regulated proteins enrichment of in grouped samples.

and 2,4-DNP degradation (Fig. 7B). A summary of significantly upregulated proteins ($p \leq 0.05$, and protein number ≥ 50) and their correlations with dominant bacterial species are shown in Supporting Information B. The aforementioned results provide a general overview of the metabolic and signaling pathways of the upregulated proteins for 2,4-DNP degradation in MBL-MHA and MHA compared with those in HA. A detailed analysis based on the KEGG pathway was subsequently conducted to improve the understanding of the process.

As shown in Fig. 7B, compared with those of anaerobic HA, protein-active pathways for ABC transporters, oxidative phosphorylation, quorum sensing (QS), biosynthesis of amino acids, and nitrogen metabolism were significantly more abundant in sample MBL-MHA-I, and oxidative phosphorylation, peroxisome, nitrogen metabolism, benzoate degradation, and the citrate cycle (i.e., TCA cycle) were more abundant in sample MBL-MHA-II ($p \leq 0.05$, and protein number ≥ 50). In addition, protein-active pathways for peroxisome, nitrogen metabolism, QS, benzoate degradation, and oxidative phosphorylation were more abundant in sample MHA than in sample anaerobic HA ($p \leq 0.05$, and protein number ≥ 50). These abundant basic pathways in MBL-MHA and MHA are primarily involved in, for example, organic matter metabolism, bioenergy transmission, and electron transfer.

For MBL-MHA, NADH-quinone oxidoreductase was the most significant oxidoreductase detected (fold change > 2.0) in the metaproteome of suspended sludge. This oxidoreductase originated from the genera *Synergistales, Propionibacterium* sp., and *Deltaproteobacteria bacterium*. NADH-quinone oxidoreductase activity has been well documented for nitro reduction processes under anaerobic conditions (Zhang et al., 2019). In addition, proteins for nitroreductase, including nitroreductase family protein and nitrate reductase subunit beta, which originated from the genera *Deltaproteobacteria* and *Actinomycetales bacteria*, were also

detected (fold change > 2.0) in the metagenome. These proteins are involved in processes that can transfer nitro groups to amino groups (Huang et al., 2012). Accordingly, in this study, aminophenol intermediates with a structure similar to nitro reduction were identified in the analysis of metabolites by LC-MS in MBL-MHA. Additionally, abundant proteins for NADH-quinone oxidoreductase and adenosine triphosphate (ATP) synthase were detected (fold change > 2.50) in the biofilm metaproteome, which originated from the gene *Brevundimonas diminuta*. According to the literature, under the micro-aeration condition, as air or oxygen has negative charges, air or oxygen could attract nicotinamide adenine dinucleotide (NAD^+) transfer, increasing the production of the reduced form of nicotinamide adenine dinucleotide (NADH) (Zhang et al., 2021). Thus, to maintain redox balance ($NADH/NAD^+$), NADH can be regenerated through NADH-quinone oxidoreductase, which produces small amounts of ATP (Kyzas et al., 2019; Zhang et al., 2021). Additionally, nitrite reductase was detected (fold change > 2.50) in biofilms that originated from the genera *Propionibacterium* sp. and *Syntrophus* sp. PtaB.Bin138, implying that nitrosation of ammonia nitrogen from suspended sludge occurred in the biofilm to generate nitrite, and the generated nitrite underwent denitrification under micro-aeration conditions (X. Mei et al., 2019; 2020). The genes annotated to pyruvate-ferredoxin/flavodoxin oxidoreductase were mainly derived (Fold change > 5.50) from cells of the genera *Propionibacterium* sp. Genes encoding succinate dehydrogenase (fold change > 7.50) and acetate kinase (fold change = 142.97) from the genera *Propionibacterium* sp. and *Brevundimonas diminuta* were significantly detected in the metagenome of the biofilm. Therefore, 2,4-DNP was supposed to be degraded to aminophenol in suspended sludge; and then further to pyruvate, succinate, and acetate; and then to the TCA metabolic process in the biofilm in MBL-MHA (X. Mei et al., 2019; Khan et al., 2019). Ammonia nitrogen, as a derivative of aminophenol degradation under micro-aeration conditions, could be oxidized to nitrite, and then the generated nitrite underwent denitrification in the biofilm (X. Mei et al., 2019; 2020).

For the MHA, proteins for NADH-quinone oxidoreductase in the detected metaproteome (fold change > 2.40) originated from the gene *Brevundimonas diminuta*, which was consistent with that in the biofilm of MBL-MHA, indicating that these NADH-quinone oxidoreductases might be primarily used to regenerate NADH rather than reduce nitro groups. Nitrite reductase was also detected (fold change > 2.00), which originated from the genera *Propionibacterium* sp. and *Actinomycetales bacterium*. However, nitroreductase was not detected sufficiently in the metaproteome, implying that the nitro groups of 2,4-DNP were directly removed in MHA. Similar results have been reported by Moreno et al. (2012) and Khan et al. (2019). Khan et al. (2019) revealed that under micro-aeration of 0.3 mg/L into an anaerobic system, 2,4-DNP was primarily degraded via the benzenethiol pathway to nitrite and benzenethiol. The genes annotated to pyruvate-ferredoxin/flavodoxin oxidoreductase were mainly derived (Fold change > 2.20) from cells of the genera *Propionibacterium* sp. Genes encoding succinate dehydrogenase from the genera *Actinomycetales* (fold change = 3.79) and *Propionibacterium* sp. (fold change = 10.28) were significantly detected in the metagenome. Therefore, 2,4-DNP was supposed to be degraded via the benzenethiol pathway to nitrite and benzenethiol, and then benzenethiol was degraded to pyruvate and succinate in MHA.

Additionally, the protein-active pathway for QS was improved under both micro-aeration HA systems, and these proteins were mainly involved in ABC transporters (Supporting Information B). Studies have reported that QS is related to the secretion of TB-EPS (Chen et al., 2019). In this study, more abundant TB-EPS was secreted in both micro-aeration HA than that in HA. These results might be the reason for the improved protein-active pathway for QS under micro-aeration conditions. In further research, the active pathway of QS proteins under micro-aeration conditions should be further investigated.

3.6. 2,4-DNP degradation mechanisms in MBL-MHA and MHA

According to the aforementioned analysis results, MBL-MHA exhibited higher 2,4-DNP degradation efficiency, higher fermentation ability, and more abundant biodiversity and richness than MHA. Considering the analyses of degradation products, EPS, microbial community, and metaproteome, the synergistic mechanisms responsible for the degradation of 2,4-DNP in MBL-MHA and MHA are proposed in Fig. 8. The underlying reasons could be mainly attributed to the following aspects: On one hand, membrane-based bubbleless micro-aeration could exhibit higher oxygen transfer ability than bubble micro-aeration, which could augment the oxygen utilization capacity and metabolic rates of microorganisms (Zhang et al., 2021). On the other hand, the redox microenvironments caused by counter-diffusion using membrane-based bubbleless micro-aeration were more beneficial for the degradation of 2,4-DNP than co-diffusion using bubble micro-aeration.

For MBL-MHA, the microenvironments caused by counter-diffusion were such that the surface layer of the biofilm and bulk liquid were anaerobic, and the remaining biofilm was micro-aerated. 2,4-DNP could be degraded first by anaerobic degradation and then by microaerobic degradation. Briefly, the corresponding aminophenols, such as 2-A-4-NP, 2-N-4-AP, and 2,4-DAP, are first formed through nitroreduction, and then phenol is formed through deamination under anaerobic conditions (Jiang et al., 2016; Tang et al., 2018). The reducing force could be formed by the expression of functional proteins such as oxidoreductase, NADP-dependent oxidoreductase, and nitroreductase, and transferred to redox proteins in EPS, which then drive nitroreduction and deamination to reduce 2,4-DNP. Additionally, nitrosation of ammonia nitrogen from suspended sludge and the denitrification process could co-occur in the biofilm under nitrite reductase. Subsequently, phenol suffered from aromatic ring cleavage to succinate and then to acetate under micro-aeration in the biofilm. The oxidation force could be formed by the expression of functional proteins, for example, succinate dehydrogenase, acetate kinase, NADH-quinone oxidoreductase, and pyruvate oxidoreductase, and transferred to redox proteins in TB-EPS, which then drive the cleavage of the aromatic ring to produce acetate or some small molecules. Thus, as more reducing and oxidizing forces became available to these reduction-oxidation related, interspecific electron transfer-related, and fermentative species in MBL-MHA, the more effective 2,4-DNP degradation performance was obtained.

For MHA, the microenvironments caused by co-diffusion were such that the surface layer of the suspended sludge and bulk liquid was microaerobic, and the inside of the sludge was anaerobic. 2,4-DNP could be degraded first by microaerobic degradation and then by anaerobic degradation. Briefly, 2,4-DNP first suffered from bond cleavage between the nitro group and the aromatic ring through hydroxylation to produce nitrite and benzenethiol. Subsequently, a small amount of benzenethiol suffered from cleavage of the aromatic ring to pyruvate and succinate. In addition, nitrite can be removed through denitrification processes. The oxidizing and reducing forces could be formed by the expression of functional proteins, for example, succinate dehydrogenase, acetate kinase, NADH-quinone oxidoreductase, pyruvate oxidoreductase, and transferred to redox proteins in EPS, which then drive the hydroxylation and nitro group removal from the benzene ring and denitrification.

Generally, denitration is the rate-limiting step of 2,4-DNP degradation under certain amounts of oxygen because of its strong electron-withdrawing property (X. Mei et al., 2019), and anaerobic reduction is conducive to its transformation to amino groups (Tang et al., 2018; X. Mei et al., 2019). Additionally, on the one hand, the toxicity of the corresponding metabolites aminophenol was significantly lower than that of 2,4-DNP (Tang et al., 2018; X. Mei et al., 2019), and on the other hand, aminophenol is more easily degraded under micro-aeration than under anaerobic conditions (X. Mei et al., 2019). Moreover, the addition of trace oxygen in an anaerobic bioreactor as a terminal electron acceptor could accelerate the hydrolysis of long-chain organic

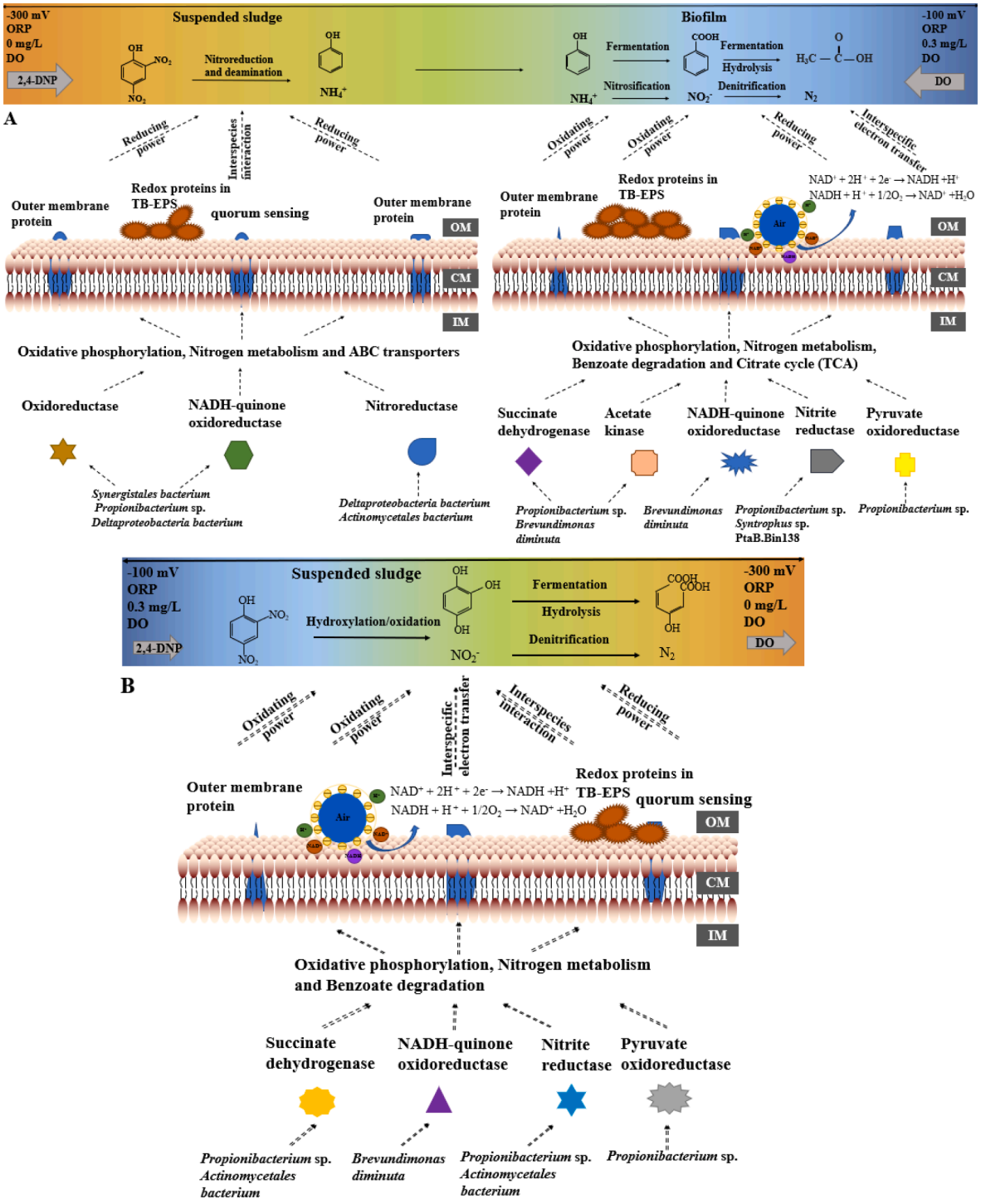


Fig. 8. Possible synergistic mechanisms responsible for the degradation 2of 2,4-DNP in MBL-MHA (A) and MHA (B).

compounds under facultative bacteria, improving the electron transfer activity of microbial metabolism (Botheju et al., 2009).

4. Conclusion

This comprehensive study examined 2,4-DNP degradation in MBL-MHA and compared it with that in MHA and anaerobic HA. The results showed that 2,4-DNP degradation could be stably enhanced in MBL-MHA and MHA during the experiment, especially in MBL-MHA. 2,4-DNP degradation rates were 84.43~97.28% in MBL-MHA, which was higher than the 82.41~94.71% in MHA under the micro-aeration of 0.5–5.0 mL air/min. These results were mainly due to the redox microenvironments caused by counter-diffusion using membrane-based bubbleless micro-aeration were more suitable for 2,4-DNP degradation than co-diffusion using bubble micro-aeration. The 2,4-DNP degradation pathways were nitroreduction, deamination, aromatic ring cleavage, and fermentation in MBL-MHA, and hydroxylation, aromatic ring cleavage, and fermentation in MHA. Abundant reduction/oxidation-related, interspecific electron transfer-related, and fermentative species were enriched in MBL-MHA. The label-free quantitative proteomic approach suggested that more reducing/oxidizing forces formed by more redox proteins/enzymes from these rich species could enhance

2,4-DNP degradation in MBL-MHA. The observed efficient, stable reactor performance demonstrated that this MBL-MHA offers substantial promise for the efficient treatment of wastewater containing nitrophenol pollutants. In further research, aspects of the fate of MBL-MHA that should be further explored are influent factors, treatment of actual wastewater containing nitrophenol, and development of new economical microaerobic membrane materials.

CRediT authorship contribution statement

Zhuowei Zhang: Methodology, Formal analysis, Writing – original draft, Data curation. **Hongbo Xi:** Formal analysis, Resources, Supervision, Writing – review & editing. **Yin Yu:** Formal analysis, Resources, Supervision, Writing – review & editing. **Changyong Wu:** Formal analysis, Resources, Supervision. **Yang Yang:** Methodology, Formal analysis. **Zhenzhen Guo:** Methodology, Formal analysis. **Yuexi Zhou:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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